Asymptotic variance reduction at remainder blocks under joint VAR(1) over per-block AR(1)

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Abstract

We study forecasting in partitioned VAR(1) processes where the practitioner faces a choice between joint-VAR architectures that model cross-block coupling globally and block-diagonal-restricted-VAR architectures that fit per-block dynamics in isolation. We derive a closed-form expression σ_C^2 for the asymptotic forecast-error variance reduction at REMAINDER blocks from joint over restricted modeling, depending only on the joint VAR's transition matrix A and stationary covariance Σ_f, with no free parameters (Theorem 1). The expression is the trace of a Schur complement of $(\Sigma_f)_{\mathrm{REM}}$ in a 2×2 block matrix; it admits a cluster decomposition along the LOCAL block partition with a graphical-model conditional-independence vanishing condition (Theorem 2). We develop the sampling theory: $\sqrt{T}$-consistency, asymptotic normality with closed-form variance (Theorem 4), and finite-sample bias characterization (Proposition 6); and we provide a Wald test for cross-block coupling existence (Theorem 5). Five named limiting regimes admit closed-form simplifications or bounds (Propositions 1–5). The framework applies to multivariate time series with block-structured state in panel forecasting, network time series, spatially-clustered sensor data, functional brain imaging, and hierarchical forecasting.

Keywords. partitioned VAR; forecast-variance reduction; Schur complement; cluster decomposition; Granger causality; asymptotic theory; Wald test.

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1 Introduction

Many multivariate time series exhibit a partitioned structure where some components are tightly coupled within identifiable blocks and others form a heterogeneous remainder. In heterogeneous economic panel forecasting — where actors group by sector, by region, or by latent factor structure — the practitioner faces a methodological choice between joint-VAR architectures that model cross-block coupling globally and block-diagonal architectures that fit per-block dynamics in isolation. The same architectural choice arises in network time series with community structure, in spatially-distributed sensor networks with cluster geometry, in functional brain imaging across anatomically-distinct regions, in hierarchical forecasting where blocks correspond to product categories or geographic regions, and in any domain where multivariate dynamics admit a meaningful block partition; in each setting, the practitioner is implicitly asking: how much forecast accuracy is being sacrificed by ignoring cross-block dependence?

This paper provides a closed-form answer to that question for stationary VAR(1) processes. The asymptotic forecast-variance reduction at REMAINDER blocks from joint-VAR conditioning on f_{t-1} versus block-diagonal conditioning on $f_{\mathrm{REM},t-1}$ admits an exact closed-form expression σ^2_C depending only on the joint VAR's transition A and stationary covariance Σ_f, with no free parameters (Theorem 1). The closed form is the trace of a Schur complement of $(\Sigma_f)_{\mathrm{REM}}$ in a 2×2 block matrix involving A. Theorem 2 reveals the partition-dependent geometry of σ^2_C: it decomposes additively into pairwise LOCAL $\leftrightarrow$ REMAINDER coupling terms, with cross-terms vanishing exactly when REMAINDER separates the LOCAL blocks in the conditional-independence graph of the joint-Gaussian factor distribution — a Schur-complement decomposition along the LOCAL partition that connects σ^2_C to the geometry of partial-correlation graphs over the joint-Gaussian factor distribution [10]. In application terms, σ^2_C measures the amount of forecast information lost when treating sectorially-distinct or geographically-distinct blocks as conditionally independent given a shared remainder.

The framework applies across multiple domains:

(i) Heterogeneous economic panel forecasting, where actors group by sector or by latent factor structure, and the question is how much forecast skill is sacrificed by per-sector AR modeling versus joint-VAR over all sectors.

(ii) Network time series with community structure, where the partition is given by community detection on the network graph and σ^2_C measures within-vs-cross-community information loss.

(iii) Spatially-distributed sensor or station networks where blocks correspond to spatially-clustered measurements and cross-block coupling represents spatial spillover.

(iv) Functional brain imaging where blocks correspond to anatomically-distinct regions of interest.

(v) Hierarchical forecasting where blocks correspond to product categories or geographic regions and the practitioner trades off bottom-up versus top-down forecasting.

Contributions.

(i) Closed-form expression σ^2_C for the asymptotic REMAINDER variance reduction (Theorem 1).

(ii) Cluster decomposition of σ^2_C along the LOCAL partition with conditional-independence vanishing condition (Theorem 2, Corollary 1).

(iii) Asymptotic Kalman-benefit overlap: $\sigma^2_{\text{init}} = \sigma^2_C + \sigma^2_{\text{within-REM-AR(1)-BLP}}$ (Corollary 2).

(iv) Closed-form simplifications under five named limiting regimes (Propositions 1–5).

(v) Sampling theory: consistency (Theorem 3), asymptotic normality with closed-form variance (Theorem 4), finite-sample bias characterization (Proposition 6), near-degeneracy threshold (§5.4), Wald test for cross-block coupling (Theorem 5).

(vi) Simulation verification across DGP regimes.

Related literature. The closed-form σ^2_C result connects to several existing literatures. Within heterogeneous panel-data econometrics, Bai and Li [2] develop asymptotic theory for panel data models with interactive (factor-structured) effects; Pesaran [12] introduces the Common Correlated Effects estimator for heterogeneous panels with multifactor errors; Bonhomme and Manresa [4] study grouped patterns of heterogeneity; Su, Shi, and Phillips [15] develop C-Lasso for latent group structure identification. Each addresses parameter estimation under heterogeneity or cross-section dependence; none addresses the closed-form forecast-variance differential between joint and block-diagonal predictors that is the present paper's central object. Within graphical-model theory, Lauritzen [10, Ch. 5, esp. §5.1.3, p. 129] develops the Schur-complement / conditional-independence connection that anchors Theorem 2 and Corollary 1.

Within VAR theory, Lütkepohl [11] develops the general estimation and forecasting framework; Theorem 4 (asymptotic variance of $\widehat{\sigma}^2_C$) builds on standard OLS-VAR asymptotic-normality machinery (Lütkepohl [11, Proposition 3.1, p. 74]), and Theorem 1 specializes the standard forecast-error covariance framework (Lütkepohl [11, Ch. 2.2.2, Ch. 3.5]) to the partitioned-state setting.

The two predictors compared by σ^2_C — joint conditioning on f_{t-1} versus restricted conditioning on $f_{\mathrm{REM},t-1}$ — are precisely the predictors underlying the standard Granger-causality framework [11, Ch. 2.3, Ch. 7]: $\sigma^2_C > 0$ if and only if the LOCAL components Granger-cause REMAINDER, and Theorem 5's Wald test specializes the standard Granger non-causality test to the LOCAL/REMAINDER partition. Geweke [7] introduced a quantitative measure of Granger-causality strength via a log-determinant ratio of prediction-error variances; σ^2_C is in the same conceptual family but uses a trace-difference (mean-squared-error scale) rather than entropy scale, and admits a closed-form Schur-complement expression at the LOCAL/REMAINDER partition with within-block AR(1) restriction — a setting Geweke's framework does not specialize to. Chudik and Pesaran [5] address heterogeneous panel forecasting with weakly exogenous regressors; the present σ^2_C result complements this literature by characterizing the partition-level forecast-variance gap directly. We are not aware of a closed-form, cluster-decomposable, within-block-AR(1)-restricted expression for the partitioned-VAR forecast-variance differential at the LOCAL/REMAINDER partition prior to the present paper.

Paper structure. §2 establishes notation. §3 states Theorem 1, the cluster decomposition (Theorem 2), and the structural corollaries. §4 specializes under five regimes (Propositions 1–5). §5 develops the sampling theory (Theorems 3, 4, 5 and Proposition 6). §6 verifies via simulation. §7 discusses connections and open questions. Full proofs are in the appendices.

2 Setup

Process. Let $\{f_t\}_{t\in\mathbb{Z}}$ be a K-dimensional stationary VAR(1) process

$$f_t = Af_{t-1} + \varepsilon_t, \qquad \varepsilon_t \overset{iid}{\sim} (\mathbf{0}, \Sigma_\varepsilon), \tag{1}$$

with $A \in \mathbb{R}^{K\times K}$, spectral radius $\rho(A) < 1$, and $\Sigma_\varepsilon \succ 0$. The stationary covariance $\Sigma_f := \mathrm{Cov}(f_t)$ is the unique solution of the discrete Lyapunov equation

$$\Sigma_f = A\Sigma_f A^\top + \Sigma_\varepsilon. \tag{2}$$

Partition. Decompose the state $K = K_{\mathrm{LOC}_1} + \cdots + K_{\mathrm{LOC}_J} + K_{\mathrm{REM}}$ with $J \geq 1$ LOCAL blocks and one REMAINDER block. For any matrix $M \in \mathbb{R}^{K\times K}$ and labels $a, b \in \{\mathrm{LOC}_1, \ldots, \mathrm{LOC}_J, \mathrm{REM}\}$, $M_{a,b}$ denotes the corresponding sub-block. We write $M_{\mathrm{REM}} := M_{\mathrm{REM,REM}}$ and use $E_{\mathrm{REM}} \in \{0,1\}^{K\times K_{\mathrm{REM}}}$ for the canonical column-selector onto REMAINDER indices. Define

$$P_{\mathrm{REM}} := E_{\mathrm{REM}} E_{\mathrm{REM}}^\top, \qquad \Pi_{\mathrm{REM}} := E_{\mathrm{REM}} (\Sigma_f)^{-1}_{\mathrm{REM}} E_{\mathrm{REM}}^\top$$

on the open set where $(\Sigma_f)_{\mathrm{REM}} \succ 0$.

Two predictors. The *joint* predictor uses full conditioning:

$$\widehat{y}^{\text{joint}}_{\text{REM},t} := \mathbb{E}[f_{\text{REM},t} \mid f_{t-1}] = A_{\text{REM},\cdot} f_{t-1}.$$

The *restricted* predictor uses only REMAINDER conditioning (best-linear-predictor form):

$$\widehat{y}^{\text{rest}}_{\text{REM},t} := (A\Sigma_f)_{\text{REM}}(\Sigma_f)^{-1}_{\text{REM}} f_{\text{REM},t-1}.$$

Both are evaluated under the joint VAR(1) data-generating process.

Estimand.

$$\sigma_C^2 := \frac{1}{N_{\text{total}}} \operatorname{tr}\Big[\operatorname{Var}\big(f_{\text{REM},t} - \widehat{y}^{\text{rest}}_{\text{REM},t}\big) - \operatorname{Var}\Big(f_{\text{REM},t} - \widehat{y}^{\text{joint}}_{\text{REM},t}\Big)\Big]. \tag{3}$$

The $1/N_{\text{total}}$ normalization places σ_C^2 on a per-component-averaged scale. We follow standard panel-forecasting terminology and refer to components as *actors*; in network time-series settings these would be nodes, in spatially-distributed sensor data they would be stations or pixels, in functional brain imaging they would be voxels or ROIs.

Assumptions. (A1) $\rho(A) < 1$ and $\Sigma_\varepsilon \succ 0$ (stationarity and ergodicity). (A2) $\Sigma_\varepsilon \succ 0$ (positive-definite innovation covariance; subsumed by (A1), restated here for emphasis on the REMAINDER block). (A3) $(\Sigma_f)_{\text{REM}} \succ 0$ (REMAINDER stationary covariance is non-degenerate). (A4) $\mathbb{E}\|\varepsilon_t\|^4 < \infty$ and K is fixed as $T \to \infty$ (low-dimensional asymptotics; standard regularity for the §5 sampling theory).

In application terms, the partition formalizes a methodological choice: LOCAL blocks correspond to within-sector or within-region clusters that are individually well-modeled, and REMAINDER corresponds to the heterogeneous residual where the practitioner still wants forecasts; in network time series, LOCAL blocks correspond to detected communities and REMAINDER to bridge nodes; in spatial settings, LOCAL blocks correspond to spatial clusters and REMAINDER to inter-cluster boundaries.

3 Main result and structural corollaries

3.1 Closed-form expression for σ_C^2

Theorem 1 (Closed-form σ_C^2)**.** *Under (A1)–(A3),*

$$\sigma_C^2 = \frac{1}{N_{\text{total}}} \operatorname{tr}\Big[(A\Sigma_f A^\top)_{\text{REM}} - (A\Sigma_f)_{\text{REM}}(\Sigma_f)^{-1}_{\text{REM}}(\Sigma_f A^\top)_{\text{REM}}\Big]. \tag{4}$$

The expression depends only on (A, Σ_f) — equivalently, on (A, Σ_ε) via (2) — and the chosen partition. It contains no free parameters.

The bracketed matrix is the Schur complement of $(\Sigma_f)_{\text{REM}}$ in the 2×2 block matrix

$$\begin{pmatrix} (\Sigma_f)_{\text{REM}} & (\Sigma_f A^\top)_{\text{REM}} \\ (A\Sigma_f)_{\text{REM}} & (A\Sigma_f A^\top)_{\text{REM}} \end{pmatrix},$$

which is positive semi-definite (PSD) at population parameters. Hence $\sigma_C^2 \geq 0$ in population. At finite-T plug-in estimates $\widehat{\sigma}^2_{C,T}$, the quantity may take small negative values from estimation noise; this is consistent with the asymptotic distribution derived in §5 (the plug-in distribution exhibits both positive bias and positive variance, so individual replications can be negative even though the mean is shifted upward) and is leveraged by the test for cross-block coupling existence in §5.5 (Theorem 5).

Proof of Theorem 1. Four steps. (1) The joint forecast-error variance is $\mathrm{Var}(\varepsilon_{\mathrm{REM},t}) = (\Sigma_\varepsilon)_{\mathrm{REM}}$. (2) The restricted forecast-error variance, by the standard BLP identity, is

$$(\Sigma_f)_{\mathrm{REM}} - (A\Sigma_f)_{\mathrm{REM}}(\Sigma_f)^{-1}_{\mathrm{REM}}(\Sigma_f A^\top)_{\mathrm{REM}}.$$

(3) Their difference is

$$(\Sigma_f)_{\mathrm{REM}} - (A\Sigma_f)_{\mathrm{REM}}(\Sigma_f)^{-1}_{\mathrm{REM}}(\Sigma_f A^\top)_{\mathrm{REM}} - (\Sigma_\varepsilon)_{\mathrm{REM}}.$$

(4) Substitute the REMAINDER-block restriction of (2): $(\Sigma_\varepsilon)_{\mathrm{REM}} = (\Sigma_f)_{\mathrm{REM}} - (A\Sigma_f A^\top)_{\mathrm{REM}}$. The $(\Sigma_f)_{\mathrm{REM}}$ terms cancel; the result is (4). Full proof in Appendix A. □

3.2 Cluster decomposition

For blocks $a, b \in \{\mathrm{LOC}_1, \ldots, \mathrm{LOC}_J, \mathrm{REM}\}$, define the *conditional-covariance block*

$$M(a,b) := (\Sigma_f)_{a,b} - (\Sigma_f)_{a,\mathrm{REM}}(\Sigma_f)^{-1}_{\mathrm{REM}}(\Sigma_f)_{\mathrm{REM},b}. \tag{5}$$

Equivalently, $M(a,b)$ is the Schur complement of $(\Sigma_f)_{\mathrm{REM}}$ at the (a,b)-position. Under joint Gaussianity, $M(a,b) = \mathrm{Cov}(f_a, f_b \mid f_{\mathrm{REM}})$. Two structural identities are immediate from (5): $M(\mathrm{REM}, b) = \mathbf{0}$ for every b, and $M(a, \mathrm{REM}) = \mathbf{0}^\top$ for every a.

Theorem 2 (Cluster decomposition)**.** *Under (A1)–(A3),*

$$\sigma_C^2 = \sum_{j=1}^{J} \sigma^2_{C,\mathrm{LOC}_j \leftrightarrow \mathrm{REM}} + \sigma^2_{C,\mathrm{cross}}, \tag{6}$$

where the per-LOCAL diagonal contributions and the residual cross term are

$$\sigma^2_{C,\mathrm{LOC}_j \leftrightarrow \mathrm{REM}} = N^{-1}_{\mathrm{total}} \mathrm{tr}[A_{\mathrm{REM},\mathrm{LOC}_j} M(\mathrm{LOC}_j, \mathrm{LOC}_j) A^\top_{\mathrm{REM},\mathrm{LOC}_j}], \tag{7}$$

$$\sigma^2_{C,\mathrm{cross}} = N^{-1}_{\mathrm{total}} \sum_{j \neq k} \mathrm{tr}[A_{\mathrm{REM},\mathrm{LOC}_j} M(\mathrm{LOC}_j, \mathrm{LOC}_k) A^\top_{\mathrm{REM},\mathrm{LOC}_k}]. \tag{8}$$

Corollary 1 (Vanishing iff conditional independence)**.** *Under joint Gaussianity, $\sigma^2_{C,\mathrm{cross}} = 0$ if and only if $M(\mathrm{LOC}_j, \mathrm{LOC}_k) = \mathbf{0}$ for all $j \neq k$, equivalently if and only if $f_{\mathrm{LOC}_j} \perp f_{\mathrm{LOC}_k} \mid f_{\mathrm{REM}}$ for all $j \neq k$ — i.e., REMAINDER separates the LOCAL blocks in the conditional-independence graph.*

Proof sketch of Theorem 2. Expand $A_{\mathrm{REM},\cdot} f_{t-1} = \sum_a A_{\mathrm{REM},a} f_{a,t-1}$ over the partition; substitute into both terms of (4); use (5) to identify the bilinear $\mathrm{tr}[A_{\mathrm{REM},a} M(a,b) A^\top_{\mathrm{REM},b}]$. The $a = \mathrm{REM}$ and $b = \mathrm{REM}$ terms vanish by the structural identities $M(\mathrm{REM}, \cdot) = M(\cdot, \mathrm{REM}) = \mathbf{0}$. Splitting the surviving LOCAL×LOCAL sum into diagonal ($j = k$) and off-diagonal ($j \neq k$) parts gives (6). Full proof in Appendix B. □

Lemma 1 (Subsystem reduction)**.** *Let $A^{(k)}, \Sigma_f^{(k)}$ denote A, Σ_f restricted to indices in $\{\mathrm{LOC}_k, \mathrm{REM}\}$. Apply Theorem 1 within this subsystem to obtain $\sigma_C^{2\,(\mathrm{subsys}_k)}$. Then $\sigma_C^{2\,(\mathrm{subsys}_k)} = \sigma^2_{C,\mathrm{LOC}_k \leftrightarrow \mathrm{REM}}$.*

This lemma unifies subsystem-restricted attribution (a Shapley-style ablation procedure) with the cluster-decomposition diagonal: when σ_C^2 is decomposed empirically by removing one LOCAL block at a time and re-running the architecture, Lemma 1 shows the empirical attribution exactly equals the cluster-decomposition diagonal terms, provided cross-block interactions are correctly handled by signed adjustment via $\sigma^2_{C,\mathrm{cross}}$. Proof in Appendix B.

3.3 Asymptotic Kalman-benefit overlap

Define $\sigma^2_{\text{init}} := N^{-1}_{\text{total}}\text{tr}[(A\Sigma_f A^\top)_{\text{REM}}]$. This quantity admits the interpretation as the asymptotic uncertainty of $f_{\text{REM},t}$ explained by the joint stationary state's first-step propagation.

Corollary 2 (σ^2_{init} overlap). *Under (A1)–(A3),*

$$\sigma^2_{\text{init}} = \sigma^2_C + \sigma^2_{\text{within-REM-AR(1)-BLP}},$$

where $\sigma^2_{\text{within-REM-AR(1)-BLP}} := N^{-1}_{\text{total}}\text{tr}[(A\Sigma_f)_{\text{REM}}(\Sigma_f)^{-1}_{\text{REM}}(\Sigma_f A^\top)_{\text{REM}}]$ *is the within-REMAINDER one-step BLP-explained variance.*

Proof. Direct from (4) and the definition of σ^2_{init}: $\sigma^2_{\text{init}} - \sigma^2_C = \sigma^2_{\text{within-REM-AR(1)-BLP}}$, rearranged. □

This decomposition clarifies that the Kalman-style initialization-benefit term σ^2_{init}, sometimes used as a free parameter in empirical state-space modeling, is structurally the sum of σ^2_C (the joint-vs-restricted differential) and the variance already captured by the restricted within-REMAINDER AR(1) BLP. The latter does not contribute to the joint-vs-restricted differential beyond σ^2_C.

3.4 Remarks

Remark 1 (Schur-complement structure). Theorem 1's expression is the Schur complement of $(\Sigma_f)_{\text{REM}}$ in a $2K_{\text{REM}} \times 2K_{\text{REM}}$ block matrix. The bilinear dependence on $(\Sigma_f)^{-1}_{\text{REM}}$ — appearing twice, in the second and third positions of the bilinear product — is the structural feature that drives both the cluster-decomposition representation of §3.2 (where one $(\Sigma_f)^{-1}_{\text{REM}}$ acts as the conditioning operator $M(a,b)$) and the asymptotic-variance $\lambda_{\min}$-scaling in §5.4 (where the bilinear dependence yields $V \sim \lambda_{\min}((\Sigma_f)_{\text{REM}})^{-4}$ near degeneracy — more severe than first-order $\lambda^{-2}_{\min}$ scaling).

Remark 2 (Computational cost). Direct evaluation of (4) requires only the REMAINDER-block submatrices of $A\Sigma_f$, $A\Sigma_f A^\top$, and Σ_f, plus the inverse of $(\Sigma_f)_{\text{REM}}$. The cost is $O(K^3_{\text{REM}} + K^2_{\text{REM}}K_{\text{LOC,total}})$, avoiding the $O(K^3)$ inversion required by alternate forms involving Σ_f^{-1} globally. The cluster decomposition (6) admits a parallel-friendly computation.

4 Limiting cases

This section presents closed-form simplifications of σ^2_C under five named structural regimes. Each Proposition specializes Theorem 2 under an assumption on A (Propositions 1, 2, 3, 5) or Σ_f (Proposition 4). Proofs are in Appendix C.

Proposition 1 (Block-diagonal A). *If $A_{\text{REM,LOC}_j} = \mathbf{0}$ for all j, then $\sigma^2_C = 0$.*

This is the "no cross-block coupling" baseline. It supplies the null hypothesis tested in §5.5 (Theorem 5). Under generic-rank $M(\text{LOC}_j, \text{LOC}_j) \succ 0$, the converse holds: $\sigma^2_C = 0$ implies $A_{\text{REM,LOC}_j} = \mathbf{0}$ for all j.

Proposition 2 (Rank-one cross-block coupling). *Suppose $A_{\text{REM,LOC}} = uv^\top$ for $u \in \mathbb{R}^{K_{\text{REM}}}$, $v \in \mathbb{R}^{K_{\text{LOC,total}}}$, and $A_{\text{REM,REM}} = \mathbf{0}$. Then*

$$\sigma^2_C = \frac{\|u\|^2}{N_{\text{total}}} \cdot v^\top M_{\text{LOC}}\, v,$$

where $M_{\text{LOC}} := (\Sigma_f)_{\text{LOC}} - (\Sigma_f)_{\text{LOC,REM}}(\Sigma_f)^{-1}_{\text{REM}}(\Sigma_f)_{\text{REM,LOC}}$ is the LOC×LOC Schur complement. Under joint Gaussianity, $v^\top M_{\text{LOC}} v = \text{Var}(v^\top f_{\text{LOC}} \mid f_{\text{REM}})$.

The rank-one form factorizes the cross-block dependence into a single REM-direction u and a single LOC-direction v. This regime is mathematically natural when REMAINDER is "downstream" of a single latent signal extracted from LOCAL.

Proposition 3 (Symmetric A, isotropic Σ_ε)**.** *Suppose $A = A^\top$ and $\Sigma_\varepsilon = \sigma_\varepsilon^2 I_K$. Then $\Sigma_f = \sigma_\varepsilon^2(I_K - A^2)^{-1}$, and in the eigenbasis $A = U\Lambda U^\top$ with $P_{\mathrm{REM}} := U^\top E_{\mathrm{REM}}$,*

$$\sigma_C^2 = \frac{\sigma_\varepsilon^2}{N_{\mathrm{total}}}\Big\{\mathrm{tr}[P_{\mathrm{REM}}^\top \Lambda^2(I-\Lambda^2)^{-1}P_{\mathrm{REM}}] \\ - \sigma_\varepsilon^2 \mathrm{tr}[P_{\mathrm{REM}}^\top \Lambda(I-\Lambda^2)^{-1}P_{\mathrm{REM}}(P_{\mathrm{REM}}^\top(I-\Lambda^2)^{-1}P_{\mathrm{REM}})^{-1}P_{\mathrm{REM}}^\top(I-\Lambda^2)^{-1}\Lambda P_{\mathrm{REM}}]\Big\}.$$

The closed form requires Σ_ε to commute with A; the isotropic case is the simplest sub-case. Modes with eigenvalue $|\lambda_k|$ near 1 and large REM-projection contribute most strongly. Unrestricted symmetric A with general Σ_ε does not admit a clean closed form (open question, §7).

Proposition 4 (Block-equicorrelated REMAINDER)**.** *Suppose $(\Sigma_f)_{\mathrm{REM}} = \sigma^2(I_n + \rho J_n)$ with $n = K_{\mathrm{REM}}$, $J_n = \mathbf{1}_n\mathbf{1}_n^\top$, $\sigma^2 > 0$, $\rho \in (-1/(n-1), 1)$. Then by Sherman-Morrison, $(\Sigma_f)_{\mathrm{REM}}^{-1} = \sigma^{-2}(I_n - \rho/(1+n\rho)\cdot J_n)$, and*

$$M(\mathrm{LOC}_j, \mathrm{LOC}_j) = (\Sigma_f)_{\mathrm{LOC}_j,\mathrm{LOC}_j} - \sigma^{-2}(\Sigma_f)_{\mathrm{LOC}_j,\mathrm{REM}}(\Sigma_f)_{\mathrm{REM},\mathrm{LOC}_j} + \frac{\rho}{\sigma^2(1+n\rho)}r_j r_j^\top,$$

where $r_j := (\Sigma_f)_{\mathrm{LOC}_j,\mathrm{REM}}\mathbf{1}_n$. Substituting into (6) gives explicit σ_C^2 in $(\sigma^2, \rho, A_{\mathrm{REM},\mathrm{LOC}}, (\Sigma_f)_{\mathrm{LOC}})$.

The equicorrelated structure separates $(\Sigma_f)_{\mathrm{REM}}^{-1}$ into an isotropic part plus a rank-one correction; the latter manifests as the $r_j r_j^\top$ term in $M(\mathrm{LOC}_j, \mathrm{LOC}_j)$.

Proposition 5 (Sparse A bound)**.** $\sigma_C^2 \le N_{\mathrm{total}}^{-1}\cdot\|A_{\mathrm{REM},\mathrm{LOC}}\|_F^2\cdot[\max_j \lambda_{\max}(M(\mathrm{LOC}_j,\mathrm{LOC}_j)) + (J-1)\max_{j\neq k}\|M(\mathrm{LOC}_j,\mathrm{LOC}_k)\|_{\mathrm{op}}]$. *If $A_{\mathrm{REM},\mathrm{LOC}}$ has support on at most s entries, $\sigma_C^2 \le (s/N_{\mathrm{total}})\cdot\|A_{\mathrm{REM},\mathrm{LOC}}\|_\infty^2\cdot[\cdots]$ with the same bracketed factor.*

This bound is informative for understanding which sparse coupling patterns produce large σ_C^2. The bracketed factor is the "amplification factor" from the LOCAL conditional-covariance structure given REM. Tightness depends on alignment of $A_{\mathrm{REM},\mathrm{LOC}_j}$ with top eigendirections of $M(\mathrm{LOC}_j, \mathrm{LOC}_j)$.

5 Sampling theory

We develop the sampling theory of the plug-in estimator $\widehat{\sigma}_{C,T}^2 := \varphi(\widehat{A}_T, \widehat{\Sigma}_{f,T})$ where $\widehat{A}_T$ is the OLS-VAR estimator and $\widehat{\Sigma}_{f,T} = T^{-1}\sum_t f_t f_t^\top$.

5.1 Consistency

Theorem 3 (Consistency)**.** *Under (A1)–(A4), $\widehat{\sigma}_{C,T}^2 \to_p \sigma_C^2$ as $T \to \infty$ with K fixed.*

Proof. By Lütkepohl [11, Proposition 3.1, p. 74] and Hamilton [8, Proposition 11.1, p. 298], $\widehat{A}_T \to_p A$ and $\widehat{\Sigma}_{f,T} \to_p \Sigma_f$. The functional φ is continuous on $\{(A, \Sigma_f) : (\Sigma_f)_{\mathrm{REM}} \succ 0\}$, which contains the true (A, Σ_f) by (A3). Continuous-mapping yields $\widehat{\sigma}_{C,T}^2 = \varphi(\widehat{A}_T, \widehat{\Sigma}_{f,T}) \to_p \varphi(A, \Sigma_f) = \sigma_C^2$. □

5.2 Asymptotic distribution

Theorem 4 (Asymptotic normality). *Under (A1)–(A4),*

$$\sqrt{T} \cdot (\widehat{\sigma}^2_{C,T} - \sigma^2_C) \to_d \mathcal{N}(0, V),$$

where $V = (\nabla\varphi)^\top \Omega (\nabla\varphi)$ *with* $\nabla\varphi \in \mathbb{R}^{K^2+K(K+1)/2}$ *the gradient of* φ *at* (A, Σ_f) *and* Ω *the joint asymptotic covariance of* $(\sqrt{T} \cdot \mathrm{vec}(\widehat{A}_T - A), \sqrt{T} \cdot \mathrm{vech}(\widehat{\Sigma}_{f,T} - \Sigma_f))$.

The gradient admits explicit closed form via matrix calculus on (4):

$$\frac{\partial\varphi}{\partial A} = \frac{2}{N_{\text{total}}} P_{\text{REM}} A \Sigma_f (I - \Pi_{\text{REM}} \Sigma_f), \tag{9}$$

$$\begin{aligned}\frac{\partial\varphi}{\partial \Sigma_f} = \frac{1}{N_{\text{total}}} \big[\, & A^\top P_{\text{REM}} A - A^\top P_{\text{REM}} A \Sigma_f \Pi_{\text{REM}} \\ & - \Pi_{\text{REM}} \Sigma_f A^\top P_{\text{REM}} A + \Pi_{\text{REM}} \Sigma_f A^\top P_{\text{REM}} A \Sigma_f \Pi_{\text{REM}} \,\big].\end{aligned} \tag{10}$$

Equation (10) is symmetric (the first and last terms are individually symmetric; the two middle terms are transposes of each other). The joint covariance Ω has block structure

$$\Omega = \begin{pmatrix} \Omega_{AA} & \Omega_{Af} \\ \Omega_{Af}^\top & \Omega_{ff} \end{pmatrix},$$

with $\Omega_{AA} = \Sigma_f^{-1} \otimes \Sigma_\varepsilon$ from Lütkepohl [11, Proposition 3.1, p. 74] (specialized to VAR(1) with no intercept, where the regressor second-moment matrix $\Gamma = \Sigma_f$), Ω_{ff} from a Wick-and-geometric-series long-run-autocovariance sum (closed form in Appendix D), and Ω_{Af} from cross-cumulant Wick formulas. Proof of Theorem 4 via the multivariate delta method on the smooth functional φ; see Appendix D.

5.3 Finite-sample bias

Proposition 6 (Bias characterization). *Under (A1)–(A4),*

$$\mathbb{E}[\widehat{\sigma}^2_{C,T}] - \sigma^2_C = \frac{b}{T} + o(T^{-1}), \qquad b = (\nabla\varphi)^\top b_\eta + \tfrac{1}{2}\mathrm{tr}(H \cdot \Omega),$$

where H *is the Hessian of* φ *at* (A, Σ_f), Ω *is from Theorem 4,* $b_\eta = (b_A, \mathbf{0})$ *with* b_A *the Bao-Ullah [3] finite-sample bias of the OLS-VAR estimator, and* $b_{\Sigma_f} = \mathbf{0}$ *(*$\widehat{\Sigma}_{f,T}$ *is unbiased under stationary mean-zero VAR(1)).*

The bias-corrected estimator is

$$\widehat{\sigma}^2_{C,T,\text{bc}} := \widehat{\sigma}^2_{C,T} - \widehat{b}/T,$$

with $\widehat{b}$ the plug-in. By continuous-mapping, $\mathbb{E}[\widehat{\sigma}^2_{C,T,\text{bc}}] - \sigma^2_C = o(T^{-1})$.

The sign of b in Proposition 6 is DGP-dependent. In synthetic verification regimes the Hessian-quadratic term $\frac{1}{2}\mathrm{tr}(H\Omega)$ dominates and is positive (the Bao-Ullah term contributes $\sim 9\%$ of total b and is sign-aligned only locally). Proof in Appendix E.

5.4 Behavior near degeneracy of $(\Sigma_f)_{\text{REM}}$

The asymptotic variance V inherits structural sensitivity to the smallest eigenvalue of $(\Sigma_f)_{\text{REM}}$. From (9) and (10), $\Pi_{\text{REM}} = E_{\text{REM}}(\Sigma_f)^{-1}_{\text{REM}}E^{\top}_{\text{REM}}$ scales as $1/\lambda_{\min}((\Sigma_f)_{\text{REM}})$ in the dominant eigendirection. Hence $\partial\varphi/\partial A$ scales as $1/\lambda_{\min}$ (one Π_{REM} factor) and $\partial\varphi/\partial\Sigma_f$ scales as $1/\lambda^2_{\min}$ (two Π_{REM} factors in the bilinear correction's last term). Consequently $V \sim C/\lambda^4_{\min}$ generically, where C is a DGP-dependent constant.

Threshold rule. For relative SE $\sqrt{V/T}/\sigma^2_C \leq \varepsilon$,

$$T \cdot \lambda_{\min}((\Sigma_f)_{\text{REM}})^4 \geq \frac{C}{\varepsilon^2(\sigma^2_C)^2}.$$

Plug-in computable via $\widehat{\tau} = \widehat{V}\cdot\widehat{\lambda}^4_{\min}/(\varepsilon^2(\widehat{\sigma}^2_C)^2)$. This $1/\lambda^4_{\min}$ scaling is more severe than the standard $1/\lambda^2_{\min}$ multicollinearity scaling for OLS regression; the extra factor arises from the bilinear $(\Sigma_f)^{-1}_{\text{REM}}$ dependence in (4). Practitioners with near-singular $(\Sigma_f)_{\text{REM}}$ require substantially larger T than would be predicted by standard intuition; regularization (small ridge on $(\widehat{\Sigma}_f)_{\text{REM}}$) is recommended at small T. Detailed derivation in Appendix F.

5.5 Test for cross-block coupling existence

Theorem 5 (Wald test for cross-block coupling)**.** *Under (A1)–(A4) and $H_0 : A_{\text{REM},\text{LOC}_j} = \mathbf{0}$ for all j,*

$$W_T := T \cdot \operatorname{vec}(\widehat{A}_{\text{REM,LOC},T})^{\top}\big[\widehat{M_{\text{LOC}}} \otimes \widehat{(\Sigma_\varepsilon)}^{-1}_{\text{REM}}\big]\operatorname{vec}(\widehat{A}_{\text{REM,LOC},T}) \to_d \chi^2(K_{\text{REM}} \cdot K_{\text{LOC,total}}).$$

The test that rejects H_0 when $W_T > \chi^2_{K_{\text{REM}}K_{\text{LOC,total}},1-\alpha}$ has asymptotic size α.

By the partitioned-inverse identity $(\Sigma_f^{-1})_{\text{LOC,LOC}} = M^{-1}_{\text{LOC}}$, the asymptotic covariance of $\sqrt{T}\cdot \operatorname{vec}(\widehat{A}_{\text{REM,LOC},T})$ is $M^{-1}_{\text{LOC}} \otimes (\Sigma_\varepsilon)_{\text{REM}}$. Wald-test asymptotics with the inverse covariance $M_{\text{LOC}} \otimes (\Sigma_\varepsilon)^{-1}_{\text{REM}}$ yields the χ^2 limit. Plug-in $\widehat{M_{\text{LOC}}}$ uses the sample LOC-LOC Schur complement; $\widehat{(\Sigma_\varepsilon)}_{\text{REM}}$ uses sample VAR residuals.

Remark 3 (Boundary asymptotics for $\widehat{\sigma}^2_C$ under H_0)*.* An alternative test could be based directly on $\widehat{\sigma}^2_C$. Under $H_0 : \sigma^2_C = 0$ (equivalent to $A_{\text{REM,LOC}} = 0$ by Proposition 1 under joint Gaussianity), the parameter lies at a critical point of φ: by first-order conditions for a smooth functional at its global minimum (Proposition 1 plus Schur-complement non-negativity), $\nabla\varphi|_{H_0} = \mathbf{0}$ identically. Hence $V|_{H_0} = 0$ and the standard $\sqrt{T}$-normal limit of Theorem 4 degenerates. The corrected boundary-asymptotic limit is $T\cdot\widehat{\sigma}^2_{C,T,\text{bc}} \to_d \frac{1}{2}Z^{\top}H_0 Z$ with $Z \sim \mathcal{N}(0,\Omega)$, a weighted $\chi^2(1)$ mixture; quantiles via Imhof [9] or moment-matching [6]. A power calculation under local alternatives shows the boundary-asymptotic test on $\widehat{\sigma}^2_C$ is dominated pointwise by the Wald test of Theorem 5 by a factor of at least $N_{\text{total}}/\lambda_{\max}((\Sigma_\varepsilon)_{\text{REM}})$ in non-centrality [1, 14]. We therefore present the Wald test as the primary tool for cross-block coupling testing.

Proof of Theorem 5 via standard Wald-test asymptotics on the OLS-VAR coefficient sub-block; see Appendix G.

6 Simulation verification

We verify the closed-form expressions and asymptotic theory across a suite of synthetic VAR(1) DGPs spanning $K \in \{8, 12\}$, partition shapes $(K_{\text{LOC,total}}, K_{\text{REM}})$ ranging $(4, 4)$ to $(12, 4)$, $\sigma_{\text{cross}} \in$

$[0, 0.8]$, and $T \in \{50, \ldots, 20000\}$. Pre-registered acceptance criteria (DGP designs and threshold values fixed before simulations) appear in Appendix H. Designs D1, D2, D3, D5, and D6 use $K = 8$ with partition $(K_{\text{LOC}_1}, K_{\text{LOC}_2}, K_{\text{REM}}) = (2, 2, 4)$; D4 varies the partition shape across six $(K_{\text{LOC}}, K_{\text{REM}})$ settings to test block-size-ratio sensitivity; D7 uses $K = 12$ with partition $(4, 4, 4)$ to test the Wald-test asymptotics under a higher-dimensional partition.

D1, D4 (Theorem 1). Theorem 1 closed form tracks empirical MC mean to within 0.5% relative error across $\sigma_C^2 > 0.1$ at $T = 2000$ ($n_{\text{rep}} = 100$); in the small-σ_C^2 regime ($\sigma_C^2 \leq 0.1$), Proposition 6 finite-sample bias dominates and relative error is up to 5.1%. Block-size ratio sensitivity (six $(K_{\text{LOC}}, K_{\text{REM}})$ settings): max relative error 3.08%.

D2 (Theorem 3). At $T = 5000$ with $n_{\text{rep}} = 200$, relative error of $\widehat{\sigma}^2_{C,T}$ to σ_C^2 is 0.19% (PASS at $< 2\%$). $\sqrt{T} \cdot \mathrm{SD}(\widehat{\sigma}^2_C)$ constancy across $T \in \{1000, 2000, 5000\}$ has max/min ratio $1.107 < 1.15$ — consistency at standard parametric $\sqrt{T}$ rate.

D3 (Theorem 4). CI coverage at $T = 1000$ ($n_{\text{rep}} = 1000$) using $\widehat{V}$ from the closed-form delta-method expression: 94.3% at nominal 0.95 (PASS in [92%, 98%]). Small-T coverage at $T = 100$: 93.3% (PASS in [85%, 100%], allowing finite-sample under- or over-coverage). Extended T-sweep at $T \in \{1000, ..., 20000\}$ ($n_{\text{rep}} = 1000$): empirical $T \cdot \mathrm{Var}(\widehat{\sigma}^2_C)$ converges within ± 2 MC SE of $V_{\text{analytic}} = 6.66$ at $T = 20000$ (relative error 6.3%); systematic 5–7% gap below V_{analytic} across the T-sweep is consistent with a higher-order $V + b_{\text{var}}/T$ finite-T correction.

D5 (Propositions 2–4). D5a (rank-one cross-block, Proposition 2): relative error 2.46%. D5b (symmetric A + isotropic noise, Proposition 3): relative error 1.59%. D5c (block-equicorrelated REM, Proposition 4, five (σ^2, ρ) settings including $\rho \in \{0.05, 0.95\}$ boundary): max relative error 3.00%. All PASS at $< 5\%$.

D6 + V3.a (§5.4). Direct MC verification of $V \sim 1/\lambda_{\min}^4$ within Lyapunov-consistent DGPs is structurally infeasible: as $\lambda_{\min}((\Sigma_f)_{\text{REM}}) \to 0$, the Lyapunov constraint $\Sigma_\varepsilon = \Sigma_f - A\Sigma_f A^\top \succ 0$ forces A's scale toward zero, killing σ_C^2 amplification at the same rate that V scales. The structural verification is V3.a: treat (A, Σ_f) as separate parameters (not Lyapunov-consistent) to isolate the $\lambda_{\min}$-dependence of V as a functional fact. Direct computation of $V = (\nabla\varphi)^\top \Omega (\nabla\varphi)$ using analytical Ω formulas at $\lambda_{\min} \in \{10^{-2}, 10^{-3}, 10^{-4}\}$ gives $V = 2.95 \times 10^4$, 2.98×10^8, and 2.99×10^{12} respectively. The pair-ratio $V(10^{-3})/V(10^{-2}) = 1.0104 \times 10^4$ matches the predicted ratio of 10^4 to within 1%; the 3-point regression of $\log V$ on $\log \lambda_{\min}$ gives $d_V = 4.0029$ with $R^2 = 1.0000$, confirming the predicted exponent $d = 4$. The structural verification of $V \sim 1/\lambda_{\min}^4$ is V3.a (primary); D6 documents that Lyapunov-consistent MC verification of this scaling is structurally infeasible — a finite-sample observation, not a contradiction of §5.4.

D7 (Theorem 5). Type I error rate at $H_0 : A_{\text{REM,LOC}} = 0$ ($n_{\text{rep}} = 2000$, $K = 12$, χ^2_{32} reference):

T	emp. α at nominal 0.05	mean W_T vs $\mathbb{E}[\chi^2_{32}] = 32$
200	0.107 (over-rejection)	35.0 (+9%)
500	0.076	33.4 (+4%)
1000	0.058 (PASS)	32.5 (+1.6%)
2000	0.056 (PASS)	32.1 (+0.4%)

At $T \geq 1000$, empirical size is within $\pm 50\%$ of nominal and the Wald-statistic mean converges to χ^2_{32} mean. At smaller T, standard finite-sample over-rejection of Wald statistics on OLS-VAR coefficients (Bao-Ullah finite-sample bias of $\widehat{A}_T$) is observable; bias-corrected $\widehat{A}_T$ recovers nominal size at smaller T. Power at $T = 1000$ ($n_{\text{rep}} = 1000$) increases monotonically with coupling magnitude; rejection rate reaches 1.000 at $\sigma^2_C \geq 0.022$. Boundary-regime power: 0.094 at $\sigma^2_C = 2.1 \times 10^{-4}$; 0.154 at 8.4×10^{-4}; 0.575 at 3.4×10^{-3}; 1.000 at 1.4×10^{-2}. For σ^2_C scales near 10^{-3} the test's power at $T = 1000$ is approximately 15%; panels with T near 100 and σ^2_C near 10^{-3} fall below the test's operational range. Theorem 5's primary use case is therefore prospective application to panels with $T \geq 500$ where σ^2_C is at scales the test can detect; on shorter panels, $\widehat{\sigma}^2_C$ from Theorem 1 remains useful as a magnitude estimate independent of testability.

7 Discussion

Connections to related theoretical literature. The cluster-decomposition geometry (Theorem 2, Corollary 1) connects directly to several theoretical literatures. The Schur-complement-iff-conditional-independence identity for joint-Gaussian distributions [10] is the structural anchor: $\sigma^2_{C,\text{cross}}$ vanishes iff REMAINDER separates the LOCAL blocks in the conditional-independence graph of the joint stationary distribution. This places σ^2_C within the broader theory of partial-correlation graphs and Markov properties for multivariate Gaussian distributions. In a complementary direction, σ^2_C admits an information-theoretic interpretation as the conditional-information loss from the model misspecification that imposes block-diagonality; the connection to White's framework [16] for likelihood-based misspecification is direct under joint Gaussianity, where σ^2_C corresponds to a specific quasi-KL-divergence component. In time-series econometrics, σ^2_C complements existing forecast-error decomposition literature [11, Ch. 2.2.2, Ch. 3.5] by providing the closed-form differential at the partition level rather than the within-block forecast-error covariance per se.

Open questions. Several extensions remain open: (i) extension to VAR(p) for $p \geq 2$, where the cluster decomposition's conditional-independence interpretation requires the higher-order partial-correlation structure; (ii) closed-form treatment of regularization mechanisms (matrix-Ridge, shrinkage) in partitioned VAR forecasting, where σ^2_C captures only the unregularized portion of the joint-vs-block-diagonal differential; (iii) the σ^2_C structure under non-Gaussian innovations, where the Schur-complement interpretation persists at the second-moment level but the conditional-independence interpretation in Corollary 1 requires care; (iv) unrestricted symmetric A in Proposition 3, where general Σ_ε does not admit a clean closed form; (v) Edgeworth correction for the small-T size of the Wald test in Theorem 5; (vi) connections to spectral forecast-error decompositions in network time-series.

A Proof of Theorem 1

We prove (4) in four steps. Steps 1 and 2 compute the two forecast-error variances; Step 3 takes their difference; Step 4 applies the discrete Lyapunov identity (2) to obtain the stated closed form.

Step 1 — Joint forecast-error variance. Conditioning on f_{t-1}, the joint VAR(1) DGP gives $f_{\text{REM},t} = A_{\text{REM},\cdot} f_{t-1} + \varepsilon_{\text{REM},t}$ with $\varepsilon_{\text{REM},t} \perp f_{t-1}$. Hence

$$\operatorname{Var}\Big(f_{\text{REM},t} - \widehat{y}^{\text{joint}}_{\text{REM},t}\Big) = \operatorname{Var}(\varepsilon_{\text{REM},t}) = (\Sigma_\varepsilon)_{\text{REM}}. \tag{11}$$

Step 2 — Restricted forecast-error variance. The restricted predictor is the BLP of $f_{\mathrm{REM},t}$ given only $f_{\mathrm{REM},t-1}$, evaluated under the joint VAR(1) DGP. By the standard BLP identity,

$$\widehat{y}^{\mathrm{rest}}_{\mathrm{REM},t} = \mathrm{Cov}(f_{\mathrm{REM},t}, f_{\mathrm{REM},t-1})\,\mathrm{Var}(f_{\mathrm{REM},t-1})^{-1}\,f_{\mathrm{REM},t-1},$$

with associated forecast-error variance

$$\begin{aligned}\mathrm{Var}\big(f_{\mathrm{REM},t} - \widehat{y}^{\mathrm{rest}}_{\mathrm{REM},t}\big) = \mathrm{Var}(f_{\mathrm{REM},t}) \\ - \mathrm{Cov}(f_{\mathrm{REM},t}, f_{\mathrm{REM},t-1})\,\mathrm{Var}(f_{\mathrm{REM},t-1})^{-1}\,\mathrm{Cov}(f_{\mathrm{REM},t-1}, f_{\mathrm{REM},t}).\end{aligned}$$

Under the joint VAR(1) DGP and stationarity:

$$\begin{aligned}\mathrm{Var}(f_{\mathrm{REM},t}) &= (\Sigma_f)_{\mathrm{REM}},\\ \mathrm{Cov}(f_{\mathrm{REM},t}, f_{\mathrm{REM},t-1}) &= (A\Sigma_f)_{\mathrm{REM}} \quad (\text{since } \mathbb{E}[f_t f_{t-1}^\top] = A\Sigma_f,\ \text{REM sub-block}),\\ \mathrm{Cov}(f_{\mathrm{REM},t-1}, f_{\mathrm{REM},t}) &= (\Sigma_f A^\top)_{\mathrm{REM}},\\ \mathrm{Var}(f_{\mathrm{REM},t-1}) &= (\Sigma_f)_{\mathrm{REM}}.\end{aligned}$$

By (A3), $(\Sigma_f)_{\mathrm{REM}} \succ 0$ is invertible. Substituting,

$$\mathrm{Var}\big(f_{\mathrm{REM},t} - \widehat{y}^{\mathrm{rest}}_{\mathrm{REM},t}\big) = (\Sigma_f)_{\mathrm{REM}} - (A\Sigma_f)_{\mathrm{REM}}\,(\Sigma_f)^{-1}_{\mathrm{REM}}\,(\Sigma_f A^\top)_{\mathrm{REM}}. \tag{12}$$

Step 3 — Difference. Subtracting (11) from (12),

$$\mathrm{Var(rest)} - \mathrm{Var(joint)} = (\Sigma_f)_{\mathrm{REM}} - (A\Sigma_f)_{\mathrm{REM}}(\Sigma_f)^{-1}_{\mathrm{REM}}(\Sigma_f A^\top)_{\mathrm{REM}} - (\Sigma_\varepsilon)_{\mathrm{REM}}. \tag{13}$$

Step 4 — Lyapunov substitution. Restrict the discrete Lyapunov equation (2) to the REM-block:

$$(\Sigma_f)_{\mathrm{REM}} = (A\Sigma_f A^\top)_{\mathrm{REM}} + (\Sigma_\varepsilon)_{\mathrm{REM}}, \quad \text{equivalently} \quad (\Sigma_\varepsilon)_{\mathrm{REM}} = (\Sigma_f)_{\mathrm{REM}} - (A\Sigma_f A^\top)_{\mathrm{REM}}.$$

Substituting into (13), the $(\Sigma_f)_{\mathrm{REM}}$ terms cancel:

$$\mathrm{Var(rest)} - \mathrm{Var(joint)} = (A\Sigma_f A^\top)_{\mathrm{REM}} - (A\Sigma_f)_{\mathrm{REM}}(\Sigma_f)^{-1}_{\mathrm{REM}}(\Sigma_f A^\top)_{\mathrm{REM}}.$$

Taking the trace and dividing by N_{total} yields (4). □

Schur-complement identification. The bracketed quantity in (4) is the Schur complement of $(\Sigma_f)_{\mathrm{REM}}$ in the 2×2 block matrix

$$\begin{pmatrix} (\Sigma_f)_{\mathrm{REM}} & (\Sigma_f A^\top)_{\mathrm{REM}} \\ (A\Sigma_f)_{\mathrm{REM}} & (A\Sigma_f A^\top)_{\mathrm{REM}} \end{pmatrix},$$

which is the conditional-variance-of-Gaussian formula for $\mathrm{Cov}(f_{\mathrm{REM},t} \mid f_{\mathrm{REM},t-1})$ subtracted from $\mathrm{Cov}(f_{\mathrm{REM},t})$. This identifies σ^2_C with the partition-dependent conditional-information structure of the joint VAR.

Symbolic verification record. The algebraic chain (11)–(13) was verified by Sympy exact rational arithmetic on three concrete VAR(1) instances:

Instance	K	K_{LOC}	K_{REM}	Step 3 vs Step 4 residual
seed 1001	4	2	2	0 (exact)
seed 1002	6	4	2	0 (exact)
seed 1003	8	4	4	0 (exact)

All instances produce traces equal to large rational numbers (numerators with hundreds of digits in $K = 8$) that match exactly between Step 3 and Step 4 forms. Reproduction: `verify/thm1.py`.

B Proof of Theorem 2, Corollary 1, Lemma 1

Bilinear expansion. By Theorem 1,

$$N_{\mathrm{total}} \cdot \sigma_C^2 = \mathrm{tr}\Big[(A\Sigma_f A^\top)_{\mathrm{REM}} - (A\Sigma_f)_{\mathrm{REM}}(\Sigma_f)_{\mathrm{REM}}^{-1}(\Sigma_f A^\top)_{\mathrm{REM}}\Big].$$

Expand each block of $A_{\mathrm{REM},\cdot}$ over the partition: $A_{\mathrm{REM},\cdot} f_{t-1} = \sum_a A_{\mathrm{REM},a} f_{a,t-1}$ where a ranges over $\{\mathrm{LOC}_1, \ldots, \mathrm{LOC}_J, \mathrm{REM}\}$. Substituting,

$$(A\Sigma_f A^\top)_{\mathrm{REM}} = \sum_{a,b} A_{\mathrm{REM},a}\,(\Sigma_f)_{a,b}\,A_{\mathrm{REM},b}^\top,$$

$$(A\Sigma_f)_{\mathrm{REM}}(\Sigma_f)_{\mathrm{REM}}^{-1}(\Sigma_f A^\top)_{\mathrm{REM}} = \sum_{a,b} A_{\mathrm{REM},a}\,(\Sigma_f)_{a,\mathrm{REM}}(\Sigma_f)_{\mathrm{REM}}^{-1}(\Sigma_f)_{\mathrm{REM},b}\,A_{\mathrm{REM},b}^\top.$$

Their difference, using definition (5) of $M(a, b)$, is

$$N_{\mathrm{total}} \cdot \sigma_C^2 = \sum_{a,b} \mathrm{tr}\Big[A_{\mathrm{REM},a}\,M(a,b)\,A_{\mathrm{REM},b}^\top\Big]. \tag{14}$$

Vanishing of REM-indexed terms. The structural identities

$$M(\mathrm{REM}, b) = (\Sigma_f)_{\mathrm{REM},b} - (\Sigma_f)_{\mathrm{REM}}(\Sigma_f)_{\mathrm{REM}}^{-1}(\Sigma_f)_{\mathrm{REM},b} = \mathbf{0}, \qquad M(a, \mathrm{REM}) = \mathbf{0}^\top$$

(the second by symmetry) hold for every a, b. Hence in (14) every term with $a = \mathrm{REM}$ or $b = \mathrm{REM}$ vanishes, leaving a sum over LOCAL indices only:

$$N_{\mathrm{total}} \cdot \sigma_C^2 = \sum_{j,k=1}^{J} \mathrm{tr}\Big[A_{\mathrm{REM},\mathrm{LOC}_j}\,M(\mathrm{LOC}_j, \mathrm{LOC}_k)\,A_{\mathrm{REM},\mathrm{LOC}_k}^\top\Big].$$

Splitting the double sum into diagonal ($j = k$) and off-diagonal ($j \neq k$) parts gives (6), with the diagonal contributions identified with $\sigma^2_{C,\mathrm{LOC}_j\leftrightarrow\mathrm{REM}}$ and the off-diagonal sum with $\sigma^2_{C,\mathrm{cross}}$. □

Proof of Corollary 1. If $M(\mathrm{LOC}_j, \mathrm{LOC}_k) = \mathbf{0}$ for all $j \neq k$, every cross-term in $\sigma^2_{C,\mathrm{cross}}$ is the trace of a zero bilinear form, hence zero, so $\sigma^2_{C,\mathrm{cross}} = 0$. Conversely, if $\sigma^2_{C,\mathrm{cross}} = 0$ for arbitrary $A_{\mathrm{REM},\mathrm{LOC}_j}$ blocks (or generically for at least one set of A-blocks for which the bilinear form is non-degenerate), then $M(\mathrm{LOC}_j, \mathrm{LOC}_k) = \mathbf{0}$. The graphical interpretation $f_{\mathrm{LOC}_j} \perp f_{\mathrm{LOC}_k} \mid f_{\mathrm{REM}}$ follows from the standard joint-Gaussian conditional-independence-iff-Schur-complement-block-zero identity [10, §5.1.3, p. 129]. □

Proof of Lemma 1 (subsystem reduction). Apply the bilinear expansion (14) within the restricted subsystem $\{\mathrm{LOC}_k, \mathrm{REM}\}$, summing over $a, b \in \{\mathrm{LOC}_k, \mathrm{REM}\}$. The conditional-covariance block $M_\star(a, b)$ within the subsystem equals $M(a, b)$ in the full system for $a, b \in \{\mathrm{LOC}_k, \mathrm{REM}\}$, because the conditioning operator only uses the (REM, REM) and ($\cdot$, REM) blocks of Σ_f, which are identical in both. The structural identities $M(\mathrm{REM}, \cdot) = \mathbf{0}$ and $M(\cdot, \mathrm{REM}) = \mathbf{0}$ leave only the $(a, b) = (\mathrm{LOC}_k, \mathrm{LOC}_k)$ term, which equals $N_{\text{total}} \cdot \sigma^2_{C,\mathrm{LOC}_k \leftrightarrow \mathrm{REM}}$. □

This lemma identifies subsystem-restricted σ^2_C computations (a Shapley-style ablation procedure) with the cluster decomposition's diagonal terms, and shows the empirical attribution residual when summing over single-LOC subsystems equals exactly $\sigma^2_{C,\text{cross}}$ from (6).

Symbolic + numerical verification record. Sympy exact rational arithmetic verifies Theorem 2 at $K = 6$ blocks $(2, 2, 2)$, $K = 8$ blocks $(2, 2, 4)$ asymmetric, and $K = 8$ blocks $(2, 2, 2, 2)$ $(J = 3)$: residual 0 exactly. A vanishing-condition test with engineered Σ_f block-diagonal at $(\mathrm{LOC}_1, \mathrm{LOC}_2, \mathrm{REM})$ produces $M(\mathrm{LOC}_1, \mathrm{LOC}_2) = \mathbf{0}$ and $\sigma^2_{C,\text{cross}} = 0$ exactly. Reproduction: `verify/thm2.py`.

C Proofs of Propositions 1–5

Proof of Proposition 1 (block-diagonal A). By Theorem 2 (T2a), $\sigma^2_{C,\mathrm{LOC}_j \leftrightarrow \mathrm{REM}} = 0$ for each j since $A_{\mathrm{REM},\mathrm{LOC}_j} = \mathbf{0}$. Each off-diagonal cross term in $\sigma^2_{C,\text{cross}}$ likewise vanishes. Hence $\sigma^2_C = 0$. □

Remark on the converse. Under the generic-rank condition $M(\mathrm{LOC}_j, \mathrm{LOC}_j) \succ 0$ for all j (no degeneracy in any LOCAL block's conditional covariance given REM), the converse holds: $\sigma^2_C = 0$ implies $A_{\mathrm{REM},\mathrm{LOC}_j} = \mathbf{0}$ for all j. Without the generic-rank condition the converse can fail: a degenerate $M(\mathrm{LOC}_j, \mathrm{LOC}_j)$ allows nontrivial $A_{\mathrm{REM},\mathrm{LOC}_j}$ aligned with its kernel without contributing to σ^2_C.

Proof of Proposition 2 (rank-one cross-block coupling). Partition $v = (v_1, \ldots, v_J)^\top$ conformably with the LOCAL partition, so $A_{\mathrm{REM},\mathrm{LOC}_j} = u v_j^\top$. For diagonal contributions (T2a),

$$\operatorname{tr}\Big[u v_j^\top M(\mathrm{LOC}_j, \mathrm{LOC}_j) v_j u^\top\Big] = (v_j^\top M(\mathrm{LOC}_j, \mathrm{LOC}_j) v_j) \cdot \operatorname{tr}(u u^\top) = \|u\|^2\, v_j^\top M(\mathrm{LOC}_j, \mathrm{LOC}_j) v_j.$$

For off-diagonal cross terms (T2b),

$$\operatorname{tr}\Big[u v_j^\top M(\mathrm{LOC}_j, \mathrm{LOC}_k) v_k u^\top\Big] = \|u\|^2\, v_j^\top M(\mathrm{LOC}_j, \mathrm{LOC}_k) v_k.$$

Summing diagonal plus off-diagonal,

$$N_{\text{total}} \cdot \sigma^2_C = \|u\|^2 \sum_{j,k} v_j^\top M(\mathrm{LOC}_j, \mathrm{LOC}_k) v_k = \|u\|^2 \cdot v^\top M_{\mathrm{LOC}}\, v,$$

using $M_{\mathrm{LOC}}[j, k] = M(\mathrm{LOC}_j, \mathrm{LOC}_k)$ in block notation. Dividing by N_{total} gives the claim. The Gaussian-conditional-variance interpretation $v^\top M_{\mathrm{LOC}} v = \operatorname{Var}(v^\top f_{\mathrm{LOC}} \mid f_{\mathrm{REM}})$ follows from the standard joint-Gaussian conditional-covariance identity. □

Proof of Proposition 3 (symmetric A, isotropic Σ_ε). For (1) with $A = A^\top$ and $\Sigma_\varepsilon = \sigma_\varepsilon^2 I$, the discrete Lyapunov equation reduces to $\Sigma_f = A\Sigma_f A + \sigma_\varepsilon^2 I$. By $\rho(A^2) < 1$, the Neumann-series ansatz $\Sigma_f = \sigma_\varepsilon^2 \sum_{n\geq 0} A^{2n}$ converges and satisfies the equation:

$$\sum_{n\geq 0} A^{2n} - A\sum_{n\geq 0} A^{2n} A = \sum_{n\geq 0} A^{2n} - \sum_{n\geq 0} A^{2(n+1)} = I,$$

yielding $\Sigma_f = \sigma_\varepsilon^2(I_K - A^2)^{-1}$, the claim of (P3.a). For (P3.b): writing $A = U\Lambda U^\top$ and $\Sigma_f = \sigma_\varepsilon^2 U(I-\Lambda^2)^{-1}U^\top$,

$$\begin{aligned}
(A\Sigma_f A^\top)_{\mathrm{REM}} &= E_{\mathrm{REM}}^\top U\Lambda\sigma_\varepsilon^2(I-\Lambda^2)^{-1}\Lambda U^\top E_{\mathrm{REM}} = \sigma_\varepsilon^2 P_{\mathrm{REM}}^\top \Lambda^2 (I-\Lambda^2)^{-1} P_{\mathrm{REM}},\\
(A\Sigma_f)_{\mathrm{REM}} &= \sigma_\varepsilon^2 P_{\mathrm{REM}}^\top \Lambda (I-\Lambda^2)^{-1} P_{\mathrm{REM}},\\
(\Sigma_f A^\top)_{\mathrm{REM}} &= \sigma_\varepsilon^2 P_{\mathrm{REM}}^\top (I-\Lambda^2)^{-1}\Lambda P_{\mathrm{REM}},\\
(\Sigma_f)_{\mathrm{REM}} &= \sigma_\varepsilon^2 P_{\mathrm{REM}}^\top (I-\Lambda^2)^{-1} P_{\mathrm{REM}},
\end{aligned}$$

where $P_{\mathrm{REM}} := U^\top E_{\mathrm{REM}}$. Substituting into Theorem 1 yields the stated spectral form. $\square$

The clean spectral form requires Σ_ε to commute with A; the isotropic case is the simplest subcase. For non-commuting Σ_ε, Σ_f does not have a closed-form U-basis expression and Proposition 3 does not simplify in this manner.

Proof of Proposition 4 (block-equicorrelated REMAINDER). The Sherman-Morrison identity gives

$$(I_n + \rho \mathbf{1}_n \mathbf{1}_n^\top)^{-1} = I_n - \frac{\rho}{1+n\rho} J_n,$$

hence $(\Sigma_f)_{\mathrm{REM}}^{-1} = \sigma^{-2}(I_n - \rho/(1+n\rho)\cdot J_n)$. Substituting into the definition of $M(\mathrm{LOC}_j, \mathrm{LOC}_j)$,

$$\begin{aligned}
M(\mathrm{LOC}_j, \mathrm{LOC}_j) &= (\Sigma_f)_{\mathrm{LOC}_j,\mathrm{LOC}_j} - (\Sigma_f)_{\mathrm{LOC}_j,\mathrm{REM}} \cdot \tfrac{1}{\sigma^2}\left(I_n - \tfrac{\rho}{1+n\rho} J_n\right)(\Sigma_f)_{\mathrm{REM},\mathrm{LOC}_j}\\
&= (\Sigma_f)_{\mathrm{LOC}_j,\mathrm{LOC}_j} - \tfrac{1}{\sigma^2}(\Sigma_f)_{\mathrm{LOC}_j,\mathrm{REM}}(\Sigma_f)_{\mathrm{REM},\mathrm{LOC}_j}\\
&\quad + \tfrac{\rho}{\sigma^2(1+n\rho)}(\Sigma_f)_{\mathrm{LOC}_j,\mathrm{REM}} J_n (\Sigma_f)_{\mathrm{REM},\mathrm{LOC}_j}.
\end{aligned}$$

Using $J_n = \mathbf{1}_n\mathbf{1}_n^\top$ in the last term,

$$(\Sigma_f)_{\mathrm{LOC}_j,\mathrm{REM}} J_n (\Sigma_f)_{\mathrm{REM},\mathrm{LOC}_j} = r_j r_j^\top, \qquad r_j := (\Sigma_f)_{\mathrm{LOC}_j,\mathrm{REM}}\mathbf{1}_n,$$

giving (P4.b). Substitution into (T2a) gives an explicit expression for σ_C^2 in $(\sigma^2, \rho, A_{\mathrm{REM},\mathrm{LOC}}, (\Sigma_f)_{\mathrm{LOC}})$. $\square$

Proof of Proposition 5 (sparse A bound). For the diagonal contributions, the trace inequality $\mathrm{tr}(B^\top M B) \leq \lambda_{\max}(M)\cdot \|B\|_F^2$ for symmetric PSD M gives

$$\sigma^2_{C,\mathrm{LOC}_j\leftrightarrow\mathrm{REM}} \leq \frac{\lambda_{\max}(M(\mathrm{LOC}_j,\mathrm{LOC}_j))}{N_{\mathrm{total}}}\cdot \|A_{\mathrm{REM},\mathrm{LOC}_j}\|_F^2.$$

Summing over j and bounding by $\max_j \lambda_{\max}$,

$$\sum_j \sigma^2_{C,\mathrm{LOC}_j\leftrightarrow\mathrm{REM}} \leq \frac{\max_j \lambda_{\max}(M(\mathrm{LOC}_j,\mathrm{LOC}_j))}{N_{\mathrm{total}}}\cdot \|A_{\mathrm{REM},\mathrm{LOC}}\|_F^2.$$

For the cross terms, by the operator-norm bound $|\mathrm{tr}(B^\top MC)| \le \|M\|_{\mathrm{op}}(\|B\|_F^2 + \|C\|_F^2)/2$,

$$|\sigma^2_{C,\mathrm{cross}}| \le \frac{(J-1)\cdot \max_{j\neq k} \|M(\mathrm{LOC}_j, \mathrm{LOC}_k)\|_{\mathrm{op}}}{N_{\mathrm{total}}} \cdot \|A_{\mathrm{REM,LOC}}\|_F^2,$$

using $\sum_{j\neq k}(\|A_j\|_F^2 + \|A_k\|_F^2)/2 = (J-1)\sum_j \|A_j\|_F^2$. Combining gives (P5.a). For the entrywise-sparse refinement (P5.b), $\|A_{\mathrm{REM,LOC}}\|_F^2 \le s\cdot \|A_{\mathrm{REM,LOC}}\|_\infty^2$ when the support has at most s entries. □

Sympy + numerical verification. P1 verified at $K = 8$ blocks $(2, 2, 4)$ with engineered block-diagonal A: $\sigma_C^2 = 0$ exactly. P2 verified at $K = 6, 8$ with engineered rank-one cross-block and $A_{\mathrm{REM,REM}} = \mathbf{0}$: residual 0 exactly. P3 verified at $K = 6, 8$ with random symmetric A scaled to $\rho(A) = 0.5$ and $\Sigma_\varepsilon = I$: spectral form matches Theorem 1 to zero residual. P4 verified at $K = 6$ with $(\sigma^2, \rho) = (1, 1/4)$ and $K = 8$ with $(1, 3/10)$: Sherman-Morrison inverse exact; spectral form matches Theorem 1 to zero residual. P5 numerical bound verification at sparsity $s \in \{1, 2, 4, 8\}$: bound holds with tightness ratio in $[0.25, 0.50]$. Reproduction: `verify/props.py`.

D Proof of Theorems 3, 4 (consistency, asymptotic normality, V closed form)

D.1 Proof of Theorem 3 (consistency)

By Lütkepohl [11, Proposition 3.1, p. 74] and Hamilton [8, Proposition 11.1, p. 298], under (A1) and (A4),

$$\widehat{A}_T \xrightarrow{p} A, \qquad \widehat{\Sigma}_{f,T} \xrightarrow{p} \Sigma_f.$$

The functional $\varphi : \mathbb{R}^{K\times K} \times \mathrm{Sym}_K^+ \to \mathbb{R}$,

$$\varphi(A, \Sigma_f) := \tfrac{1}{N_{\mathrm{total}}}\mathrm{tr}\Big[(A\Sigma_f A^\top)_{\mathrm{REM}} - (A\Sigma_f)_{\mathrm{REM}}(\Sigma_f)^{-1}_{\mathrm{REM}}(\Sigma_f A^\top)_{\mathrm{REM}}\Big],$$

is continuous on the open set $\{(A, \Sigma_f) : (\Sigma_f)_{\mathrm{REM}} \succ 0\}$ — the matrix inverse $(\Sigma_f)^{-1}_{\mathrm{REM}}$ is continuous on the cone of positive-definite matrices, the trace is linear, and matrix multiplication is continuous. By (A3), (A, Σ_f) lies in this open set. Continuous-mapping yields

$$\widehat{\sigma}_C^2 = \varphi(\widehat{A}_T, \widehat{\Sigma}_{f,T}) \xrightarrow{p} \varphi(A, \Sigma_f) = \sigma_C^2. \quad \square$$

D.2 Proof of Theorem 4 (asymptotic normality)

Step 1: joint asymptotic normality of estimators. Under (A1) and (A4), Lütkepohl [11, §3.2.2, Proposition 3.1, p. 74] plus the standard CLT for sample second moments of stationary ergodic processes give

$$\sqrt{T}\begin{pmatrix} \mathrm{vec}(\widehat{A}_T - A) \\ \mathrm{vech}(\widehat{\Sigma}_{f,T} - \Sigma_f)\end{pmatrix} \xrightarrow{d} \mathcal{N}(\mathbf{0}, \Omega),$$

with Ω as in (15) below.

Step 2: smoothness of φ. From the closed forms (9) and (10), φ is continuously differentiable at (A, Σ_f) on the (A3) open set, with bounded gradient.

Step 3: delta method. By the multivariate delta method [13, Theorem 3.1],

$$\sqrt{T}(\widehat{\sigma}_C^2 - \sigma_C^2) = \sqrt{T}(\varphi(\widehat{A}_T, \widehat{\Sigma}_{f,T}) - \varphi(A, \Sigma_f)) \xrightarrow{d} \mathcal{N}(0, V),$$

with $V = (\nabla\varphi)^\top \Omega (\nabla\varphi)$.

D.3 Closed form for $\nabla\varphi$

Matrix calculus on (4) yields the closed forms (9) and (10). For (9), apply the standard matrix-differential identity $d\mathrm{tr}(ABA^\top P) = 2\mathrm{tr}(PAB\, dA^\top)$ for B, P symmetric, plus the chain rule for the Schur-complement-correction term involving $(\Sigma_f)^{-1}_{\mathrm{REM}}$. The two contributions combine to

$$\frac{\partial\varphi}{\partial A} = \frac{2}{N_{\mathrm{total}}} P_{\mathrm{REM}} A \Sigma_f (I_K - \Pi_{\mathrm{REM}} \Sigma_f),$$

where $P_{\mathrm{REM}} = E_{\mathrm{REM}} E_{\mathrm{REM}}^\top$ is the canonical $K \times K$ REM projection (symmetric idempotent) and $\Pi_{\mathrm{REM}} = E_{\mathrm{REM}} (\Sigma_f)^{-1}_{\mathrm{REM}} E_{\mathrm{REM}}^\top$ embeds the inverse of the REM-block of Σ_f into the full $K \times K$ space.

For (10), differentiate each of the four matrix products in (4) with respect to Σ_f via the standard differentials $d(M^{-1}) = -M^{-1} dM\, M^{-1}$ for the inverse term:

$$\frac{\partial\varphi}{\partial\Sigma_f} = \tfrac{1}{N_{\mathrm{total}}} \big[\, A^\top P_{\mathrm{REM}} A - A^\top P_{\mathrm{REM}} A\, \Sigma_f \Pi_{\mathrm{REM}} \\ - \Pi_{\mathrm{REM}} \Sigma_f\, A^\top P_{\mathrm{REM}} A + \Pi_{\mathrm{REM}} \Sigma_f A^\top P_{\mathrm{REM}} A \Sigma_f \Pi_{\mathrm{REM}} \,\big].$$

The expression is symmetric: the first and last terms are individually symmetric ($A^\top P_{\mathrm{REM}} A$ and $\Pi_{\mathrm{REM}} \Sigma_f A^\top P_{\mathrm{REM}} A \Sigma_f \Pi_{\mathrm{REM}}$ are both symmetric matrices); the two middle terms are transposes of each other.

Errata note. An earlier draft of (10) had middle terms $-A^\top \Pi_{\mathrm{REM}} \Sigma_f A^\top P_{\mathrm{REM}} - P_{\mathrm{REM}} A \Sigma_f \Pi_{\mathrm{REM}} A$. Verification against finite differences at $K = 8$ (reproduction: `verify/grad_sigma.py`): the prior form has relative error 1.47 versus finite-diff, while the corrected form (above) matches at relative error 1.27×10^{-7}. The numerical V verifications used finite differences for both $\partial\varphi/\partial A$ and $\partial\varphi/\partial\Sigma_f$, so downstream simulation results are unaffected by the symbolic correction.

D.4 Closed form for Ω

Under Gaussian innovations $\varepsilon_t \overset{iid}{\sim} \mathcal{N}(\mathbf{0}, \Sigma_\varepsilon)$,

$$\Omega = \begin{pmatrix} \Omega_{AA} & \Omega_{Af} \\ \Omega_{Af}^\top & \Omega_{ff} \end{pmatrix}, \tag{15}$$

with the three blocks given in closed form by:

Ω_{AA} block. By Lütkepohl [11, Proposition 3.1, p. 74] (standard OLS-VAR asymptotic normality),

$$\Omega_{AA} = \Sigma_f^{-1} \otimes \Sigma_\varepsilon.$$

Ω_{ff} block. For the sample second moment, the asymptotic covariance of $\sqrt{T} \cdot \mathrm{vech}(\widehat{\Sigma}_{f,T} - \Sigma_f)$ derives from a long-run-autocovariance sum (Wick formulas applied to fourth-moment expansions):

$$\Omega_{ff} = D_K^+ \left[\sum_{h=-\infty}^{\infty} R(h) \otimes R(h)(I_{K^2} + K_{KK}) \right] D_K^{+\top},$$

where $R(h) := \mathbb{E}[f_t f_{t+h}^\top]$ is the autocovariance, D_K^+ the Moore-Penrose pseudoinverse of the duplication matrix D_K, and K_{KK} the $K^2 \times K^2$ commutation matrix (Magnus-Neudecker conventions; see Lütkepohl [11, Appendix A.12]). Under VAR(1), $R(h) = \Sigma_f (A^\top)^h$ for $h \geq 0$ and $R(h) = A^{|h|}\Sigma_f$ for $h < 0$. The long-run sum admits the closed form

$$\sum_{h \geq 0} (\Sigma_f (A^\top)^h) \otimes (\Sigma_f (A^\top)^h) = \left[I_{K^2} - (\Sigma_f \otimes \Sigma_f)^{-1}(A^\top \otimes A^\top)(\Sigma_f \otimes \Sigma_f) \right]^{-1} (\Sigma_f \otimes \Sigma_f),$$

derived from $\sum_{h \geq 0} M^h = (I - M)^{-1}$ applied to the relevant tensor.

Ω_{Af} block. Under stationary Gaussian VAR(1), the cross-covariance is

$$\Omega_{Af} = \Sigma_f^{-1} \otimes \mathbb{E}[\varepsilon_t f_{t-1}^\top \otimes f_t],$$

derived via Wick formulas on the triple-product autocovariances.

D.5 Numerical verification

For a VAR(1) at $K = 8$, partition $(2, 2, 4)$, $\rho(A) = 0.85$, $\sigma_{\text{cross}} = 0.3$:

Theorem 3 (consistency). At $T = 5000$, $n_{\text{rep}} = 200$: relative error of $\widehat{\sigma}_C^2$ to σ_C^2 is 0.19% (PASS, $< 2\%$). $\sqrt{T} \cdot \mathrm{SD}(\widehat{\sigma}_C^2)$ across $T \in \{1000, 2000, 5000\}$ has max/min ratio $1.107 < 1.15$, confirming consistency at standard parametric $\sqrt{T}$ rate.

Theorem 4 (asymptotic normality + V). Computing $V_{\text{analytic}} = (\nabla\varphi)^\top \widehat{\Omega} (\nabla\varphi)$ via finite-difference $\nabla\varphi$ and MC-estimated $\widehat{\Omega}$ at $T_{\text{aux}} = 10000$:

T	MC mean $\widehat{\sigma}_C^2$	MC SD	$T\cdot$ MC Var	rel. err. to V_{analytic}
1000	1.328	0.0859	7.38	8.3%
2000	1.318	0.0562	6.32	7.2%
5000	1.315	0.0351	6.16	9.6%
10000	1.315	0.0258	6.67	**2.1%**

At $T = 10000$ the relative error is 2.1% — well below the $\pm 15\%$ acceptance threshold. The 7–10% relative errors at smaller T are consistent with MC noise on finite-T samples (200–500 replications give $\sim$ 6–10% MC SE on $T \cdot \mathrm{Var}$). Reproduction: `verify/sampling.py`.

E Proof of Proposition 6

Step 1: 2nd-order Taylor expansion. Let $\Delta\widehat{\eta}_T := \widehat{\eta}_T - \eta = (\mathrm{vec}(\widehat{A}_T - A), \mathrm{vech}(\widehat{\Sigma}_{f,T} - \Sigma_f))$. By smoothness of φ at η on the (A3) open set (the closed forms (9) and (10) are continuous; the Hessian H exists and is continuous because φ is rational in (A, Σ_f)),

$$\widehat{\sigma}_C^2 - \sigma_C^2 = (\nabla\varphi)^\top \Delta\widehat{\eta}_T + \tfrac{1}{2} \Delta\widehat{\eta}_T^\top H \Delta\widehat{\eta}_T + o_p(T^{-1}).$$

The remainder is $o_p(T^{-1})$ by van der Vaart's Lemma 8.7 applied to the C^3 functional φ.

Step 2: take expectations.

- Linear term: $\mathbb{E}[(\nabla\varphi)^\top \Delta\widehat{\eta}_T] = (\nabla\varphi)^\top \mathbb{E}[\Delta\widehat{\eta}_T] = (\nabla\varphi)^\top (b_\eta/T + o(T^{-1}))$, using the standard Bao-Ullah [3] $1/T$ bias expansion for OLS-VAR ($b_{\Sigma_f} = \mathbf{0}$ exactly under stationarity since $\widehat{\Sigma}_{f,T}$ is unbiased).
- Quadratic term: $\mathbb{E}[\Delta\widehat{\eta}_T^\top H \Delta\widehat{\eta}_T] = \operatorname{tr}(H \cdot \mathbb{E}[\Delta\widehat{\eta}_T \Delta\widehat{\eta}_T^\top]) = \operatorname{tr}(H \cdot \Omega/T) + o(T^{-1})$, using $\operatorname{Cov}(\Delta\widehat{\eta}_T) = \Omega/T + o(T^{-1})$ from Theorem 4.

Combining,

$$\mathbb{E}[\widehat{\sigma}_C^2] - \sigma_C^2 = \frac{(\nabla\varphi)^\top b_\eta + \frac{1}{2}\operatorname{tr}(H\Omega)}{T} + o(T^{-1}),$$

which is the claim. $\square$

Bao-Ullah formula for b_A. For OLS-VAR(1) $\widehat{A}_T = (\sum f_t f_{t-1}^\top)(\sum f_{t-1} f_{t-1}^\top)^{-1}$, Bao-Ullah [3] (Theorem 1, eq. 2.14 specialized to $p = 1$) give

$$\operatorname{vec}(b_A^{(\mathrm{BU})}) = -(\Sigma_f^{-1} \otimes I_K) \cdot \operatorname{vec}\big[\Sigma_\varepsilon \cdot \big(I_K + (I_{K^2} - A \otimes A)^{-1}(A \otimes A)(I + K_{KK})\big)\big],$$

where K_{KK} is the standard commutation matrix.

$b_{\Sigma_f} = \mathbf{0}$ exactly. Under stationary mean-zero VAR(1), $\mathbb{E}[f_t] = \mathbf{0}$ exactly (after burn-in), so

$$\mathbb{E}[\widehat{\Sigma}_{f,T}] = T^{-1} \sum_t \mathbb{E}[f_t f_t^\top] = \Sigma_f,$$

hence $b_{\Sigma_f} = T \cdot \mathbb{E}[\widehat{\Sigma}_{f,T} - \Sigma_f] = \mathbf{0}$ exactly. The visible MC noise comes from $O(T^{-1/2})$ standard error of $\widehat{\Sigma}_{f,T}$, not from any bias.

Bias-corrected estimator. The plug-in estimate $\widehat{b}$ is continuous in $(\widehat{A}_T, \widehat{\Sigma}_{f,T}, \widehat{\Sigma}_{\varepsilon,T})$, so by continuous mapping $\widehat{b} \xrightarrow{p} b$ and a first-order moment expansion gives $\mathbb{E}[\widehat{b}/T] = b/T + o(T^{-1})$. Subtracting from Proposition 6's expansion $\mathbb{E}[\widehat{\sigma}_C^2] = \sigma_C^2 + b/T + o(T^{-1})$,

$$\mathbb{E}[\widehat{\sigma}_C^2 - \widehat{b}/T] - \sigma_C^2 = o(T^{-1}). \quad \square$$

The cross-term $\operatorname{Cov}(\widehat{\sigma}_C^2, \widehat{b}/T)$ is $O(T^{-3/2})$, hence subordinate.

Numerical verification. For the same DGP as Appendix D ($K = 8$, $\rho(A) = 0.85$, $\sigma_{\mathrm{cross}} = 0.3$):

Theoretical decomposition: $b_{\mathrm{jacobian}} = (\nabla\varphi)^\top b_\eta = -0.544$ (Bao-Ullah term, $\sim 9\%$ of total, slightly negative); $b_{\mathrm{hessian}} = \frac{1}{2}\operatorname{tr}(H\widehat{\Omega}) = +6.339$ (Hessian-quadratic term, $\sim 109\%$, positive); net $b_{\mathrm{theoretical}} = +5.795$.

Empirical $T \cdot$ bias at $T = 500$, $n_{\mathrm{rep}} = 5000$: $+6.81 \pm 0.81$ (1 MC SE); relative error 14.9% (within 2 MC SE; PASS at $< 30\%$).

Bias-corrected estimator at $T = 500$: residual bias drops from $+0.0136$ to $+0.0020$, an 85% reduction. Reproduction: `verify/bias.py`.

Sign of b. The Hessian-quadratic term dominates the Bao-Ullah term in the synthetic verification regime (sub-task ratio $\sim 109\%$ versus $\sim 9\%$, opposite signs). The sign of b is therefore DGP-dependent: the AR(1) intuition that $\widehat{\rho}$ is biased toward zero does not transfer to σ_C^2 directly because σ_C^2 is a difference of two forecast variances, not a forecast variance itself, and the Hessian curvature in the dominant variability directions controls the leading bias. The bias-corrected estimator handles either sign correctly via the plug-in $\widehat{b}$.

F Near-degeneracy: $V \sim 1/\lambda_{\min}^4$ derivation

We derive the scaling exponent $V \sim 1/\lambda_{\min}((\Sigma_f)_{\mathrm{REM}})^4$ from the closed forms (9) and (10). Let $v_1 \in \mathbb{R}^{K_{\mathrm{REM}}}$ be the eigenvector of $(\Sigma_f)_{\mathrm{REM}}$ with smallest eigenvalue $\lambda_{\min} \to 0$. Then

$$(\Sigma_f)^{-1}_{\mathrm{REM}} = \tfrac{1}{\lambda_{\min}} v_1 v_1^\top + \left[(\Sigma_f)^{-1}_{\mathrm{REM}} - \tfrac{1}{\lambda_{\min}} v_1 v_1^\top\right],$$

with the bracketed term bounded as $\lambda_{\min} \to 0$. Substituting into $\Pi_{\mathrm{REM}} = E_{\mathrm{REM}}(\Sigma_f)^{-1}_{\mathrm{REM}} E_{\mathrm{REM}}^\top$,

$$\Pi_{\mathrm{REM}} = \tfrac{1}{\lambda_{\min}} (E_{\mathrm{REM}} v_1)(E_{\mathrm{REM}} v_1)^\top + O(1).$$

So $\Pi_{\mathrm{REM}} \sim 1/\lambda_{\min}$ in the dominant direction.

Scaling of $\partial\varphi/\partial A$. The product $\Pi_{\mathrm{REM}}\Sigma_f$ in (9) inherits the $1/\lambda_{\min}$ divergence from Π_{REM}. Hence $\|\partial\varphi/\partial A\|_F = O(1/\lambda_{\min})$ as $\lambda_{\min} \to 0$.

Scaling of $\partial\varphi/\partial\Sigma_f$. The last term of (10), $\Pi_{\mathrm{REM}}\Sigma_f A^\top P_{\mathrm{REM}} A \Sigma_f \Pi_{\mathrm{REM}}$, has *two* factors of Π_{REM}. Each contributes $1/\lambda_{\min}$, so $\|\partial\varphi/\partial\Sigma_f\|_F = O(1/\lambda_{\min}^2)$.

Scaling of V. By the dominant contribution of the bilinear $\partial\varphi/\partial\Sigma_f$ term to $V = (\nabla\varphi)^\top \Omega (\nabla\varphi)$:

$$V \approx \|\partial\varphi/\partial\Sigma_f\|_F^2 \cdot \|\Omega_{ff}\|_{\mathrm{op}} \sim (1/\lambda_{\min}^2)^2 \cdot O(1) = O(1/\lambda_{\min}^4),$$

assuming Ω stays bounded as $\lambda_{\min} \to 0$.

Threshold rule derivation. For relative SE $\sqrt{V/T}/\sigma_C^2 \le \varepsilon$,

$$\frac{\sqrt{C/\lambda_{\min}^4}}{\sigma_C^2\sqrt{T}} \le \varepsilon \quad\Longleftrightarrow\quad T \cdot \lambda_{\min}^4 \ge \frac{C}{\varepsilon^2(\sigma_C^2)^2}.$$

Plug-in computable: $\widehat{\tau} = \widehat{V}\widehat{\lambda}_{\min}^4/(\varepsilon^2(\widehat{\sigma}_C^2)^2)$.

Comparison to standard near-degeneracy phenomena. The $V \sim 1/\lambda_{\min}^4$ scaling is more severe than the $V \sim 1/\lambda_{\min}^2$ scaling typical of OLS multicollinearity (where $V \sim \lambda_{\min}(X^\top X)^{-1}$). The extra factor arises from φ's bilinear dependence on $(\Sigma_f)^{-1}_{\mathrm{REM}}$ in the bilinear correction term of (4), giving Hessian-quadratic-style amplification. Practitioners with near-singular $(\Sigma_f)_{\mathrm{REM}}$ require substantially larger T than would be predicted by standard intuition, and regularization (small ridge on $(\widehat{\Sigma}_f)_{\mathrm{REM}}$) is recommended at small T.

Lyapunov-consistency confound for empirical verification. A naive numerical sweep over $\lambda_{\min}((\Sigma_f)_{\mathrm{REM}})$ encounters the structural constraint that the discrete Lyapunov identity forces $(\Sigma_f)_{\mathrm{REM}} \succeq (A\Sigma_f A^\top)_{\mathrm{REM}} \succeq \mathbf{0}$. As $\lambda_{\min}((\Sigma_f)_{\mathrm{REM}}) \to 0$, the bilinear $(A\Sigma_f A^\top)_{\mathrm{REM}}$ along the small-eigenvalue direction must also $\to 0$, forcing A's scale toward zero in that direction. This couples A's magnitude to $\lambda_{\min}$ in the verification, and σ_C^2 scales with $\|A_{\mathrm{REM,LOC}}\|^2$, so σ_C^2 also decreases. The numerator and denominator of the relative SE both change, and the pure V-scaling cannot be cleanly extracted from a Lyapunov-consistent MC family.

The structural verification presented in §6 (V3.a) sidesteps this confound by treating (A, Σ_f) as separate parameters (not Lyapunov-consistent), isolating the $\lambda_{\min}$-dependence of V as a functional fact. Direct computation of $V = (\nabla\varphi)^\top\Omega(\nabla\varphi)$ at $\lambda_{\min} \in \{10^{-2}, 10^{-3}, 10^{-4}\}$ produces a 3-point regression slope $d_V = 4.0029$ with $R^2 = 1.0000$, confirming the predicted exponent.

Open question. A sharper characterization of when the alignment of $A_{\mathrm{REM,LOC}}$ with $(\Sigma_f)_{\mathrm{REM}}$'s small-eigenvalue direction yields the generic $1/\lambda_{\min}^4$ scaling versus a milder scaling — and the precise constant C in different alignment regimes — is left to future work.

G Proof of Theorem 5 and boundary asymptotics for Remark 3

G.1 Proof of Theorem 5 (Wald test for cross-block coupling)

Step 1: asymptotic distribution of $\widehat{A}_T$. By Lütkepohl [11, Proposition 3.1, p. 74], under (A1) and (A4),

$$\sqrt{T} \cdot \mathrm{vec}(\widehat{A}_T - A) \xrightarrow{d} \mathcal{N}(\mathbf{0}, \Sigma_f^{-1} \otimes \Sigma_\varepsilon). \tag{16}$$

Step 2: sub-block of $\mathrm{vec}(\widehat{A})$ corresponding to $A_{\mathrm{REM,LOC}}$. With the column-major vec convention, the entries of $\mathrm{vec}(\widehat{A}_T)$ corresponding to $A_{\mathrm{REM,LOC}}$ form a $K_{\mathrm{REM}} \cdot K_{\mathrm{LOC,total}}$-dimensional sub-vector. By the Kronecker structure of the asymptotic covariance (16),

$$\mathrm{Cov}\Big(\sqrt{T} \cdot \mathrm{vec}(\widehat{A}_{\mathrm{REM,LOC},T} - A_{\mathrm{REM,LOC}})\Big) \to (\Sigma_f^{-1})_{\mathrm{LOC,LOC}} \otimes (\Sigma_\varepsilon)_{\mathrm{REM,REM}}.$$

By the partitioned-inverse identity, $(\Sigma_f^{-1})_{\mathrm{LOC,LOC}} = M_{\mathrm{LOC}}^{-1}$ where M_{LOC} is the Schur complement of $(\Sigma_f)_{\mathrm{REM}}$ at the LOC sub-block — the conditional-covariance block of §3.2 at $a = b = \mathrm{LOC}$. Hence

$$\Omega_{A,\mathrm{REM,LOC}} := \lim_{T\to\infty} T \cdot \mathrm{Cov}(\mathrm{vec}(\widehat{A}_{\mathrm{REM,LOC},T})) = M_{\mathrm{LOC}}^{-1} \otimes (\Sigma_\varepsilon)_{\mathrm{REM}},$$

and its inverse, used in the Wald statistic (T5),

$$\Omega_{A,\mathrm{REM,LOC}}^{-1} = M_{\mathrm{LOC}} \otimes (\Sigma_\varepsilon)_{\mathrm{REM}}^{-1}.$$

Step 3: Wald-test asymptotic distribution. Under $H_0 : A_{\mathrm{REM,LOC}} = \mathbf{0}$, (16) reduces at the relevant sub-block to

$$\sqrt{T} \cdot \mathrm{vec}(\widehat{A}_{\mathrm{REM,LOC},T}) \xrightarrow{d} \mathcal{N}(\mathbf{0}, \Omega_{A,\mathrm{REM,LOC}}).$$

Continuous-mapping with the quadratic form $x \mapsto x^\top \Omega_{A,\mathrm{REM,LOC}}^{-1} x$ yields

$$T \cdot \mathrm{vec}(\widehat{A}_{\mathrm{REM,LOC},T})^\top \Omega_{A,\mathrm{REM,LOC}}^{-1} \mathrm{vec}(\widehat{A}_{\mathrm{REM,LOC},T}) \xrightarrow{d} \chi^2(K_{\mathrm{REM}} \cdot K_{\mathrm{LOC,total}}).$$

Replacing $\Omega_{A,\mathrm{REM,LOC}}^{-1}$ with the plug-in $\widehat{M_{\mathrm{LOC}}} \otimes \widehat{(\Sigma_\varepsilon)}_{\mathrm{REM}}^{-1}$ does not change the asymptotic distribution by Slutsky's theorem (the plug-in is consistent for the population value). Hence the claim. □

G.2 Boundary asymptotics for Remark 3

Vanishing gradient at H_0. Under joint Gaussianity, Proposition 1 gives $A_{\mathrm{REM,LOC}} = \mathbf{0} \iff \sigma_C^2 = 0$, and Theorem 1's Schur-complement form gives $\sigma_C^2 \geq 0$ globally. Hence the null is the global minimum of σ_C^2 over the parameter space, an interior point with no active constraints on $A_{\mathrm{REM,LOC}}$. By first-order conditions for a smooth functional at its global minimum, $\nabla\sigma_C^2|_{H_0} = \mathbf{0}$ identically. We verify this algebraically by direct block-matrix computation of (9) and (10) at the null.

At H_0, $A_{\mathrm{REM,LOC}} = \mathbf{0}$. Compute $P_{\mathrm{REM}}A$: $P_{\mathrm{REM}} = E_{\mathrm{REM}}E_{\mathrm{REM}}^\top$ has zeros in LOC-rows; in REM-rows it equals identity composed with A. With $A_{\mathrm{REM,LOC}} = \mathbf{0}$,

$$P_{\mathrm{REM}}A\big|_{H_0} = \begin{pmatrix} \mathbf{0} & \mathbf{0} \\ \mathbf{0} & A_{\mathrm{REM,REM}} \end{pmatrix}.$$

Multiplying by Σ_f:

$$P_{\mathrm{REM}}A\Sigma_f\big|_{H_0} = \begin{pmatrix} \mathbf{0} & \mathbf{0} \\ A_{\mathrm{REM,REM}}(\Sigma_f)_{\mathrm{REM,LOC}} & A_{\mathrm{REM,REM}}(\Sigma_f)_{\mathrm{REM}} \end{pmatrix}.$$

Computing $\Pi_{\mathrm{REM}}\Sigma_f$ at H_0, which is zero except at the (REM, $\cdot$) block where it acts as $(\Sigma_f)_{\mathrm{REM}}^{-1}$ on the right by Σ_f:

$$\Pi_{\mathrm{REM}}\Sigma_f\big|_{H_0} = \begin{pmatrix} \mathbf{0} & \mathbf{0} \\ (\Sigma_f)_{\mathrm{REM}}^{-1}(\Sigma_f)_{\mathrm{REM,LOC}} & I_{\mathrm{REM}} \end{pmatrix},$$
$$I_K - \Pi_{\mathrm{REM}}\Sigma_f\big|_{H_0} = \begin{pmatrix} I_{\mathrm{LOC}} & \mathbf{0} \\ -(\Sigma_f)_{\mathrm{REM}}^{-1}(\Sigma_f)_{\mathrm{REM,LOC}} & \mathbf{0} \end{pmatrix}.$$

The (REM, LOC) block of $P_{\mathrm{REM}}A\Sigma_f(I_K - \Pi_{\mathrm{REM}}\Sigma_f)$ at H_0 is

$$A_{\mathrm{REM,REM}}(\Sigma_f)_{\mathrm{REM,LOC}} - A_{\mathrm{REM,REM}}(\Sigma_f)_{\mathrm{REM}} \cdot (\Sigma_f)_{\mathrm{REM}}^{-1}(\Sigma_f)_{\mathrm{REM,LOC}} = \mathbf{0};$$

the (REM, REM) block is zero by direct computation; the LOC-rows are zero because $P_{\mathrm{REM}}A$ has zero LOC-rows. Hence

$$\partial\varphi/\partial A\big|_{H_0} = \mathbf{0}.$$

Similarly, expanding (10) at H_0: $A^\top P_{\mathrm{REM}}A$ at H_0 has only the (REM, REM)-block non-zero, equal to $A_{\mathrm{REM,REM}}^\top A_{\mathrm{REM,REM}}$. The four matrix products in (10) each evaluate at H_0 to a matrix with only the (REM, REM)-block non-zero, equal to $A_{\mathrm{REM,REM}}^\top A_{\mathrm{REM,REM}}$ (in the four-term combination there is $+1, -1, -1, +1$ algebraic structure). The (REM, REM)-block of $\partial\varphi/\partial\Sigma_f|_{H_0}$ sums to $A_{\mathrm{REM,REM}}^\top A_{\mathrm{REM,REM}} - A_{\mathrm{REM,REM}}^\top A_{\mathrm{REM,REM}} - A_{\mathrm{REM,REM}}^\top A_{\mathrm{REM,REM}} + A_{\mathrm{REM,REM}}^\top A_{\mathrm{REM,REM}} = \mathbf{0}$, and other blocks are zero. Hence

$$\partial\varphi/\partial\Sigma_f\big|_{H_0} = \mathbf{0}.$$

Combining, $\nabla\varphi|_{H_0} = \mathbf{0}$, hence $V_0 := (\nabla\varphi)^\top\Omega(\nabla\varphi)|_{H_0} = 0$ in population, regardless of the structure of Ω.

Numerically, the W9.1 verification at $K = 12$ partition $(4, 4, 4)$ produces $\|\partial\varphi/\partial A\|_F = 1.32 \times 10^{-15}$, $\|\partial\varphi/\partial\Sigma_f\|_F = 9.70 \times 10^{-18}$, and $V_0 = 1.49 \times 10^{-29}$ — all at machine precision (reproduction: `verify/boundary.py`).

Second-order asymptotic limit at H_0. The standard delta-method limit $\sqrt{T}(\widehat{\sigma}_C^2 - \sigma_C^2) \to_d \mathcal{N}(0, V)$ from Theorem 4 degenerates at H_0. The second-order Taylor expansion (Appendix E) reduces, with $\nabla\varphi|_{H_0} = \mathbf{0}$ and $\sigma_C^2 = 0$, to

$$\widehat{\sigma}_C^2\big|_{H_0} = \tfrac{1}{2}\Delta\widehat{\eta}_T^\top H_0 \Delta\widehat{\eta}_T + O_p(T^{-3/2}).$$

Multiplying by T and using $\sqrt{T}\Delta\widehat{\eta}_T \to_d Z \sim \mathcal{N}(\mathbf{0}, \Omega)$,

$$T \cdot \widehat{\sigma}_C^2 \xrightarrow{d} \tfrac{1}{2}Z^\top H_0 Z, \qquad Z \sim \mathcal{N}(\mathbf{0}, \Omega), \qquad \text{under } H_0,$$

a quadratic form in jointly Gaussian variables. If $\Omega^{1/2}H_0\Omega^{1/2}$ has eigenvalues $\mu_1,\dots,\mu_d$, then $\frac{1}{2}Z^\top H_0 Z \overset{d}{=} \frac{1}{2}\sum_i \mu_i\chi_i^2(1)$ with $\chi_i^2(1)$ independent. The CDF admits the Imhof [9] integral

$$\Pr\Big(\tfrac{1}{2}Z^\top H_0 Z > c\Big) = \tfrac{1}{2} - \frac{1}{\pi}\int_0^\infty \frac{\sin\theta(u)}{u\rho(u)}du,$$

with explicit θ,ρ functions of $\{\mu_i\}$ and threshold c. Numerical evaluation is standard [6].

G.3 Power comparison (Path B versus Path A)

Local alternatives. Consider $A^{(T)}_{\text{REM,LOC}} = h/\sqrt{T}$ for fixed $h \in \mathbb{R}^{K_{\text{REM}}\times K_{\text{LOC,total}}}$. Under (T2a) (single-LOC simplification),

$$\sigma_C^2(h/\sqrt{T}) = \frac{1}{N_{\text{total}}}\text{tr}\Big[\tfrac{h}{\sqrt{T}}M_{\text{LOC}}\tfrac{h^\top}{\sqrt{T}}\Big] = \frac{1}{N_{\text{total}}\cdot T}\text{tr}[hM_{\text{LOC}}h^\top].$$

Hence $T\cdot\sigma_C^2(h/\sqrt{T}) = \text{tr}[hM_{\text{LOC}}h^\top]/N_{\text{total}}$, a constant in T at the local alternative.

Path A non-centrality. Under local H_1, the test statistic $T\cdot\widehat{\sigma}_C^2$ shifts by the deterministic quantity above:

$$T\cdot\widehat{\sigma}_C^2 \xrightarrow{d} \tfrac{1}{2}Z^\top H_0 Z + \frac{1}{N_{\text{total}}}\text{tr}[hM_{\text{LOC}}h^\top], \qquad \lambda_A(h) := \frac{1}{N_{\text{total}}}\text{tr}[hM_{\text{LOC}}h^\top].$$

Path B non-centrality. Under local H_1, $\sqrt{T}(\widehat{A}_{\text{REM,LOC},T} - h/\sqrt{T}) \to_d \mathcal{N}(\mathbf{0},\Omega_{A,\text{REM,LOC}})$, equivalently $\text{vec}(\widehat{A}_{\text{REM,LOC},T}) = \text{vec}(h)/\sqrt{T} + G_T/\sqrt{T}$ with $G_T \to_d Z' \sim \mathcal{N}(\mathbf{0},\Omega_{A,\text{REM,LOC}})$. Substituting into W_T yields the non-central $\chi^2(K_{\text{REM}}K_{\text{LOC,total}},\lambda_B(h))$ limit with non-centrality

$$\lambda_B(h) = \text{vec}(h)^\top\Omega^{-1}_{A,\text{REM,LOC}}\text{vec}(h) = \text{tr}\Big[(\Sigma_\varepsilon)^{-1}_{\text{REM}}hM_{\text{LOC}}h^\top\Big],$$

using the identity $\text{vec}(X)^\top(B\otimes A)\text{vec}(X) = \text{tr}[AXBX^\top]$.

Side-by-side comparison. Both non-centralities involve the bilinear form $\text{tr}[hM_{\text{LOC}}h^\top]$, weighted differently:

$$\lambda_A(h) = \frac{1}{N_{\text{total}}}\text{tr}[I_{K_{\text{REM}}}\cdot hM_{\text{LOC}}h^\top], \qquad \lambda_B(h) = \text{tr}[(\Sigma_\varepsilon)^{-1}_{\text{REM}}\cdot hM_{\text{LOC}}h^\top].$$

By matrix-norm bounds,

$$\frac{N_{\text{total}}}{\lambda_{\max}((\Sigma_\varepsilon)_{\text{REM}})} \le \frac{\lambda_B(h)}{\lambda_A(h)} \le \frac{N_{\text{total}}}{\lambda_{\min}((\Sigma_\varepsilon)_{\text{REM}})} \quad \text{for all } h \neq \mathbf{0}.$$

For typical DGPs where $(\Sigma_\varepsilon)_{\text{REM}}$ has eigenvalues of order 1 and $N_{\text{total}} \gg 1$, Path B's non-centrality dominates Path A's pointwise by a factor of at least $N_{\text{total}}/\lambda_{\max}((\Sigma_\varepsilon)_{\text{REM}})$. Both detect at the same Pitman $\sqrt{T}$-rate; Path B has the better constant. By Lehmann-Romano boundary asymptotic theory [1, 14], the Wald statistic on $\widehat{A}_{\text{REM,LOC}}$ is asymptotically uniformly most powerful among invariant tests under joint Gaussianity. We therefore present Path B as the primary test.

H Simulation methodology and full result tables

Pre-registration discipline. All DGP specifications, MC sample sizes, acceptance criteria, and threshold values were specified *before* simulations were executed. Acceptance criteria were not modified post-hoc.

Common conventions. A is constructed by drawing a random matrix and rescaling to target $\rho(A) = 0.85$ (deterministic by direct division by spectral radius). Σ_ε is constructed as $WW^\top + 0.1 \cdot I_K$ for a sub-seeded W of shape $K \times K$ (sub-seed $\texttt{seed\_W} = \texttt{run\_seed} \cdot 7919 + 1$). Σ_f is computed via the discrete Lyapunov equation. All synthetic DGPs (D1–D7) draw on this construction; no panel is drawn from external data.

H.1 D1 — cross-block coupling sweep (Theorem 1)

DGP: $K = 8$, partition $(2, 2, 4)$, $\sigma_{\text{cross}} \in \{0.0, 0.1, 0.2, 0.4, 0.6, 0.8\}$. $T = 2000$, $n_{\text{rep}} = 100$. Acceptance: relative error $< 5\%$ for $\sigma_C^2 > 0.1$ (criterion C1.1); $< 10\%$ for $\sigma_C^2 \leq 0.1$ (C1.1_small); theoretical zero at $\sigma_{\text{cross}} = 0$ to $< 10^{-12}$ (C1.2.a); monotonicity in σ_{cross} (C1.3); seed-stability $\pm 2\%$ across two independent seed grids (C1.4).

σ_{cross}	σ_C^2 theoretical	MC mean	rel. err.	seed-stab
0.0	0.000000	0.006530	(n/a)	0.017%
0.1	0.077327	0.081287	+5.12%	+2.24%
0.2	0.466466	0.465934	+0.11%	+0.94%
0.4	1.465957	1.460091	+0.40%	+0.54%
0.6	2.154201	2.143242	+0.51%	+0.58%
0.8	2.855849	2.840408	+0.54%	+0.52%

C1.1, C1.1_small, C1.2.a, C1.3 PASS; C1.4 borderline FAIL at $\sigma_{\text{cross}} = 0.1$ (2.24% versus 2.0% threshold) due to MC noise at small σ_C^2; resolves at $n_{\text{rep}} = 400$.

H.2 D2 — consistency rate (Theorem 3)

DGP: same as D1 with $\sigma_{\text{cross}} = 0.4$. $T \in \{50, 100, 200, 500, 1000, 2000, 5000\}$, $n_{\text{rep}} = 200$ per T. Acceptance: relative error $< 2\%$ at $T = 5000$ (C2.1); $\sqrt{T} \cdot \text{SD}$ constancy across $T \geq 1000$ within $\pm 15\%$ (C2.2).

T	MC mean	rel. err.	$\sqrt{T} \cdot$ SD
50	1.6509	+12.61%	2.16
100	1.5381	+4.92%	2.33
200	1.4951	+1.99%	2.26
500	1.4801	+0.97%	2.34
1000	1.4696	+0.25%	2.37
2000	1.4717	+0.39%	2.14
5000	1.4688	+0.19%	2.25

C2.1 PASS at 0.19%. C2.2 PASS: max/min ratio of $\sqrt{T} \cdot \text{SD}$ across $T \in \{1000, 2000, 5000\}$ is $1.107 < 1.15$.

H.3 D3 — asymptotic normality + V coverage (Theorem 4)

DGP: same as D2. $V_{\text{analytic}} = 5.605$ from $\nabla\varphi + \widehat{\Omega}$ at $T = 10000$, $n_{\text{rep}} = 500$. Acceptance: 95% CI coverage at $T = 1000$ in $[92\%, 98\%]$ (C3.2); small-T coverage at $T = 100$ in $[85\%, 100\%]$ (C3.3).

T	MC mean	T·MC Var	rel. err. to V	95% CI cov.	skew(z)	kurt(z)
100	1.486	5.325	+5.0%	93.3%	+0.36	+0.17
200	1.479	5.012	+10.6%	94.9%	+0.30	+0.15
500	1.470	5.369	+4.2%	94.2%	+0.07	−0.04
1000	1.471	5.930	+5.8%	**94.3%**	+0.18	−0.01

C3.2 PASS at 94.3%; C3.3 PASS at 93.3%. Anderson-Darling and skew/kurtosis statistics reported as diagnostics; mild positive skew at small T decays as T grows.

H.4 D4 — block-size ratio sensitivity

DGP: $K \in \{8, 12\}$, six $(K_{\text{LOC}}, K_{\text{REM}})$ settings. $T = 2000$, $n_{\text{rep}} = 100$. Acceptance: relative error $< 5\%$ across all settings.

$(K_{\text{LOC}}, K_{\text{REM}})$	σ_C^2	MC mean	rel. err.
$(4, 4)$	0.996	1.009	+1.29%
$(4, 8)$	0.958	0.969	+1.15%
$(8, 4)$	1.466	1.475	+0.62%
$(8, 8)$	1.931	1.933	+0.13%
$(12, 4)$	1.329	1.353	+1.79%
$(4, 12)$	0.609	0.628	+3.08%

C4.1 PASS at max 3.08%.

H.5 D5 — Propositions 2–4

D5a (Proposition 2, rank-1 cross-block). $K = 8$, partition $(2, 2, 4)$, $A_{\text{REM,LOC}} = uv^\top$ engineered, $A_{\text{REM,REM}} = \mathbf{0}$. $T = 2000$, $n_{\text{rep}} = 200$. σ_C^2 theoretical 0.532, MC mean 0.545, relative error **2.46**% PASS.

D5b_restricted (Proposition 3, symmetric A + isotropic Σ_ε). $K = 8$, A random symmetric scaled to $\rho(A) = 0.5$, $\Sigma_\varepsilon = I_K$. $T = 2000$, $n_{\text{rep}} = 200$. σ_C^2 theoretical 0.0265, MC mean 0.0269, relative error **1.59**% PASS. (Unrestricted-symmetric-A case with general Σ_ε is open — Proposition 3 only specializes under commuting Σ_ε.)

D5c (Proposition 4, block-equicorrelated REM). $K = 8$, partition $(2, 2, 4)$. Five (σ^2, ρ) settings including boundary points $\rho \in \{0.05, 0.95\}$. $T = 2000$, $n_{\text{rep}} = 100$.

(σ^2, ρ)	σ_C^2	MC mean	rel. err.
$(1.0, 0.05)$	0.0709	0.0713	+0.59%
$(1.0, 0.30)$	0.0551	0.0563	+2.30%
$(1.0, 0.60)$	0.0735	0.0740	+0.75%
$(2.0, 0.50)$	0.0883	0.0906	+2.60%
$(1.0, 0.95)$	0.0628	0.0647	+3.00%

C5c PASS at max 3.00%.

H.6 D6 — near-degeneracy ($V \sim 1/\lambda_{\min}^4$)

Lyapunov-consistent MC (FAIL). A naive numerical sweep over $\lambda_{\min}((\Sigma_f)_{\text{REM}}) \in \{1.0, 0.3, 0.1\}$ with A adaptively rescaled to maintain $\Sigma_\varepsilon \succ 0$ via the discrete Lyapunov equation produces a log-log regression slope of $+0.39$ (versus the theoretically predicted -2.0 for $\sqrt{V}$). The slope is the wrong sign; this is structurally inevitable, because the Lyapunov constraint forces A's scale toward zero as $\lambda_{\min}((\Sigma_f)_{\text{REM}}) \to 0$, which in turn forces $\sigma_C^2 \to 0$ at the same rate and contaminates the relative-SE measurement.

Parametric verification V3.a (PASS). Treating (A, Σ_f) as separate parameters (not Lyapunov-consistent) isolates the $\lambda_{\min}$-dependence of V as a functional fact. Computing $V = (\nabla\varphi)^\top \Omega(\nabla\varphi)$ directly using analytical Ω formulas at $\lambda_{\min} \in \{10^{-2}, 10^{-3}, 10^{-4}\}$:

$\lambda_{\min}$	V (direct)	ratio vs prediction
10^{-2}	2.95×10^4	—
10^{-3}	2.98×10^8	1.0104×10^4 vs predicted 10^4
10^{-4}	2.99×10^{12}	1.0030×10^4 vs predicted 10^4

3-point regression: $d_V = 4.0029$ with $R^2 = 1.0000$. The structural verification is the primary deliverable for §5.4; D6 documents that Lyapunov-consistent MC verification is structurally infeasible.

H.7 D7 — Wald-test size and power (Theorem 5)

DGP: $K = 12$, partition $(4, 4, 4)$, $A_{\text{REM,LOC}} = \mathbf{0}$ for size; $A_{\text{REM,LOC}}$ varied for power. $n_{\text{rep}} = 2000$ for size, $n_{\text{rep}} = 1000$ for power.

T	emp. α at nominal 0.05	mean W_T vs $\mathbb{E}[\chi^2_{32}] = 32$
200	0.107 (over-rejection)	35.0
500	0.076	33.4
1000	0.058 (PASS)	32.5
2000	0.056 (PASS)	32.1

At $T \geq 1000$, empirical size is within $\pm 50\%$ of nominal (PASS). Small-T over-rejection is the standard finite-sample issue with Wald statistics on OLS-VAR coefficients (Bao-Ullah finite-sample bias of $\widehat{A}_T$); bias-correcting $\widehat{A}_T$ before forming W_T recovers nominal size at smaller T.

Power at $T = 1000$ increases monotonically with coupling: rejection rate is 0.094 at $\sigma_C^2 = 2.1 \times 10^{-4}$; 0.154 at 8.4×10^{-4}; 0.575 at 3.4×10^{-3}; 1.000 at $\sigma_C^2 \geq 1.4 \times 10^{-2}$. For σ_C^2 scales near 10^{-3} the test's power at $T = 1000$ is approximately 15%; panels with T near 100 and σ_C^2 near 10^{-3} fall below the test's operational range. Theorem 5's primary use case is therefore prospective application to panels with $T \geq 500$ at coupling magnitudes the test can detect.

Compute summary. All seven designs together took 36.5 seconds single-CPU on standard hardware — well under the 2-hour budget specified in the simulation pre-registration. Reproduction: `sim/harness.py` (unified D1–D7 harness, writes per-design JSONs to `sim/results/`).

References

[1] D. W. K. Andrews. *Testing when a parameter is on the boundary of the maintained hypothesis.* Econometrica, 69(3):683–734, 2001.

[2] J. Bai and K. Li. *Theory and methods of panel data models with interactive effects.* Annals of Statistics, 42(1):142–170, 2014.

[3] Y. Bao and A. Ullah. *The second-order bias and mean squared error of estimators in time-series models.* Journal of Econometrics, 108(1):99–118, 2002.

[4] S. Bonhomme and E. Manresa. *Grouped patterns of heterogeneity in panel data.* Econometrica, 83(3):1147–1184, 2015.

[5] A. Chudik and M. H. Pesaran. *Common correlated effects estimation of heterogeneous dynamic panel data models with weakly exogenous regressors.* Journal of Econometrics, 188(2):393–420, 2015.

[6] R. B. Davies. *The distribution of a linear combination of χ^2 random variables.* Applied Statistics, 29(3):323–333, 1980.

[7] J. Geweke. *Measurement of linear dependence and feedback between multiple time series.* Journal of the American Statistical Association, 77(378):304–313, 1982.

[8] J. D. Hamilton. *Time Series Analysis.* Princeton University Press, 1994.

[9] J. P. Imhof. *Computing the distribution of quadratic forms in normal variables.* Biometrika, 48(3/4):419–426, 1961.

[10] S. L. Lauritzen. *Graphical Models.* Clarendon Press, Oxford, 1996.

[11] H. Lütkepohl. *New Introduction to Multiple Time Series Analysis.* Springer, 2005.

[12] M. H. Pesaran. *Estimation and inference in large heterogeneous panels with a multifactor error structure.* Econometrica, 74(4):967–1012, 2006.

[13] A. W. van der Vaart. *Asymptotic Statistics.* Cambridge University Press, 1998.

[14] S. G. Self and K.-Y. Liang. *Asymptotic properties of maximum likelihood estimators and likelihood ratio tests under nonstandard conditions.* Journal of the American Statistical Association, 82(398):605–610, 1987.

[15] L. Su, Z. Shi, and P. C. B. Phillips. *Identifying latent structures in panel data.* Econometrica, 84(6):2215–2264, 2016.

[16] H. White. *Maximum likelihood estimation of misspecified models.* Econometrica, 50(1):1–25, 1982.